

# Submission pack

ROLL & DEAL · FOUR ORIGINAL TABLE GAMES · VERSION 0.1.0

*Dice rolled by the player. Cards dealt by the dealer.*

**Dice Spread** — *Will the card land between your dice?* Main wager Between, RTP 96.30%. [Rules of Play](#) · [Math Report](#), revision 3, 2026-09-30.

**Moving Target** — *Your dice set the target. Will the cards land on it?* Main wager Exact Hit, RTP 96.15%. [Rules of Play](#) · [Math Report](#), revision 3, 2026-09-30.

**Mirror** — *Your dice against the dealer's cards, hand for hand.* Main wager Mirror, RTP 94.91%. [Rules of Play](#) · [Math Report](#), revision 3, 2026-09-30.

**Lock & Roll** — *Lock a die, roll the other, beat the cards.* Main wager Lock & Roll, RTP 95.93%. [Rules of Play](#) · [Math Report](#), revision 3, 2026-09-30.

|           |                                                                                   |
|-----------|-----------------------------------------------------------------------------------|
| Author    | Jonathas Costa                                                                    |
| Code      | commit 47bab22a941c983cd3e18fd6989ebb7a1e6479b5 (2026-09-30)                      |
| Figures   | docs/results.json, from each game's mathSummary() and its tests' recorded results |
| Generator | tools/generate-docs.ts (pnpm docs:generate)                                       |
| Demo      | https://jonathascosta.github.io/CasinoGames/                                      |

For each game, the **Rules of Play** are for a pit manager or a live-dealer trainer: the equipment, the layout, the wagers, the procedure, the settlement and the rulings, with no figure beyond the payable. The **Math Report** is for a gaming-lab mathematician or a distributor's analyst: every wager's return, derived exactly and confirmed by seeded simulation, each table naming the test that reproduces it. To reproduce every figure: clone the repository, run `pnpm install`, `pnpm test` (exact figures), `pnpm test:math` (simulations, 124,852,059 rounds for Dice Spread alone) and `pnpm docs:check`; each fails if its figures differ from those recorded here.

# Dice Spread — Rules of Play

Will the card land between your dice?

## 1. Game and document

| Game       | Dice Spread, an original table game of the Roll & Deal family                                                                 |
|------------|-------------------------------------------------------------------------------------------------------------------------------|
| Version    | 0.1.0                                                                                                                         |
| Document   | Rules of Play, revision 3                                                                                                     |
| Date       | 2026-09-30                                                                                                                    |
| Author     | Jonathas Costa                                                                                                                |
| Companion  | Math Report ( docs/math/dice-spread.md )                                                                                      |
| Demo table | <a href="https://jonathascosta.github.io/CasinoGames/dice-spread">https://jonathascosta.github.io/CasinoGames/dice-spread</a> |

### Revision history

| Revision | Date       | Author         | Change                                                                                                                                                                                         |
|----------|------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | 2026-09-30 | Jonathas Costa | First game sheet: rules, declared math, exact and Monte Carlo tests. Card counting exposure measured.                                                                                          |
| 2        | 2026-09-30 | Jonathas Costa | Renamed from its working title, with English names for the game and its bets.                                                                                                                  |
| 3        | 2026-09-30 | Jonathas Costa | Reformatted for submission as two documents, Rules of Play and Math Report, generated from the code with every figure traced to the engine or to a test; card counting measured on every game. |

Money is written in the table's currency with two decimals, as the demo shows it (the demo plays with virtual chips).

## 2. Objective

The player rolls two dice and the dealer deals one card from a shoe of cards ace to 6. The main wager, Between, wins when the card falls strictly between the two dice and pays more the closer together they are; optional side wagers settle on the same roll and card.

## 3. Equipment

- **Dice.** Two standard six-sided dice, faces 1 to 6, thrown by the player (by the round's shooter at a live table: see section nine). The **spread** of a roll is the higher die minus the lower one: 0 for a pair, 5 for a 1 and a 6.
- **Shoe.** 6 decks, each of the A to the 6 in 4 suits (24 cards): 144 cards in all, made from standard decks with every card above the 6 removed. The ace counts 1 and every other card its number.
- **Cut card.** It goes in with 36 cards behind it, a penetration of 75%: 108 cards are dealt before it comes out. When it comes out, the round in progress is completed, and the whole shoe is shuffled before the next round.
- **Cosmetic.** Suits never count: no wager looks at a card's suit, only at its value. Which die is which, and the order in which they stop, do not matter either: a roll is read by its two values.
- **Also.** The layout (section four), chips, a discard rack and, at a live table, a dice cup or shaker.

## 4. Table layout

Each player position has one spot for the main wager, Between, in front, and one for each side wager around it: Match, Bullseye, Doubles and Triple. The felt prints what each spot pays:

| Spot                   | Wins when                                                   | Pays    | A 1.00 wager wins |
|------------------------|-------------------------------------------------------------|---------|-------------------|
| Between, spread 2      | the card strictly between the dice (1 value)                | 4 to 1  | 4.00              |
| Between, spread 3      | the card strictly between the dice (2 values)               | 2 to 1  | 2.00              |
| Between, spread 4      | the card strictly between the dice (3 values)               | 1 to 1  | 1.00              |
| Between, spread 5      | the card strictly between the dice (4 values)               | 1 to 2  | 0.50              |
| Between, spread 0 or 1 | no card can fall between the dice                           | push    | stake returned    |
| Match                  | the card equals either die                                  | 2 to 1  | 2.00              |
| Bullseye               | the dice are 2 apart and the card is the value between them | 22 to 1 | 22.00             |
| Doubles                | the dice are a pair                                         | 4 to 1  | 4.00              |
| Triple                 | the dice are a pair and the card has their value            | 30 to 1 | 30.00             |

The demo shows this layout at <https://jonathascosta.github.io/CasinoGames/dice-spread>. At the top are the dice and the dealer's card, then a strip of the values 1 to 6 that marks the dice and lights the values between them. Below it is the felt: Between in the centre with the side wagers around it, and Between's pays by spread printed along the bottom, the spread in force lit. Under the felt sit the chips, the balance, the total wagered and the Clear, Roll and Autoplay buttons, with Rules and Paytable buttons beside the balance; an RTP monitor below the table compares the rounds played with the declared figures.

## 5. Wagers

Every wager is placed before the roll and settled once the card is turned. The main wager comes first; the side wagers follow in the felt's order. The game has no progressive wager.

### 5.1 Between (main wager)

- **When.** Before the roll. It can be placed alone, and every side wager needs it.
- **Limits.** 0.50 to 250.00.

- **Wins** when the card's value lies strictly between the two dice: higher than the lower die and lower than the higher one.
- **Pays** by the spread: 4 to 1 on a spread of 2, 2 to 1 on a spread of 3, 1 to 1 on a spread of 4 and 1 to 2 on a spread of 5.
- **Pushes** when the dice are a pair or one apart (spread 0 or 1): no card can fall between them, so the wager is returned whatever the card.
- **Loses** when the spread is 2 or more and the card equals either die or lies outside them.

## 5.2 Match (side wager)

- **When.** Before the roll, only with a Between wager.
- **Limits.** 0.50 to 25.00.
- **Wins** when the card equals either die(it pays once, even when both dice show the card's value).
- **Pays** 2 to 1.
- **Pushes** never.
- **Loses** on any other roll and card.

## 5.3 Bullseye (side wager)

- **When.** Before the roll, only with a Between wager.
- **Limits.** 0.50 to 25.00.
- **Wins** when the dice are 2 apart and the card is the value between them (dice 6-4 and a 5, say).
- **Pays** 22 to 1.
- **Pushes** never.
- **Loses** on any other roll and card.

## 5.4 Doubles (side wager)

- **When.** Before the roll, only with a Between wager.
- **Limits.** 0.50 to 25.00.
- **Wins** when the dice are a pair, whatever the card; it is settled with the other wagers once the card is turned.
- **Pays** 4 to 1.
- **Pushes** never.
- **Loses** on any other roll and card.

## 5.5 Triple (side wager)

- **When.** Before the roll, only with a Between wager.
- **Limits.** 0.50 to 25.00.
- **Wins** when the dice are a pair and the card has their value.
- **Pays** 30 to 1.
- **Pushes** never.
- **Loses** on any other roll and card.

## 6. Sequence of play

1. **Wagers.** The dealer invites wagers. Each player places Between and any side wagers, within the limits. The dealer then closes betting ("No more bets").
2. **Shuffle, if due.** If the cut card came out during the previous round, the dealer shuffles the whole shoe, has it cut and places the cut card with 36 cards behind it before the round begins.
3. **Roll.** The player rolls both dice. The dealer reads the roll aloud with its spread and the Between pay in force: "5 and 2, spread 3, Between pays 2 to 1".
4. **Card.** The dealer deals one card from the shoe, face down, then turns it face up. No other card is dealt.
5. **Compare.** The dealer compares the card with the dice (section seven) and announces each wager's result.
6. **Settle.** The dealer settles every wager (section eight), clears the card to the discard rack and returns the dice for the next round.

## 7. Comparing the card with the dice

There are no hands to rank: the one card is compared with the two dice.

- A card is **between** the dice when its value is higher than the lower die and lower than the higher die. A card equal to either die is not between them.
- The **spread** is the higher die minus the lower die; a pair has a spread of 0.
- A card **matches** a die when it has the same value.

Worked examples:

- Dice 5-2 (spread 3), card 3: Between wins 2 to 1; Match, Bullseye, Doubles and Triple lose.
- Dice 6-4 (spread 2), card 5: Between wins 4 to 1; Bullseye wins 22 to 1; Match, Doubles and Triple lose.
- Dice 3-3 (spread 0), card 3: Match wins 2 to 1; Doubles wins 4 to 1; Triple wins 30 to 1; Between pushes; Bullseye loses.
- Dice 6-1 (spread 5), card 6: Match wins 2 to 1; Between, Bullseye, Doubles and Triple lose.

## 8. Settlement procedure

- The dealer settles once the card is turned: first collecting every losing wager, then paying the winners, Between first and the side wagers in the felt's order (Match, Bullseye, Doubles and Triple), and leaving pushed wagers in place.
- A winning wager is paid at its odds and keeps its stake. The table above shows what a 1.00 wager wins on each line.

- A pay that is not a whole number of cents (1 to 2 on an odd number of cents) is rounded down to the cent.
- There is no fee, commission or progressive meter.

## 9. Physical and live-dealer adaptation

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- **Dice.** The player throws from a dice cup or by hand, across the layout, so that both dice tumble and come to rest on the felt. At a table with several players, one shooter throws per round, in turn, and that one roll and one card settle every seat; an automatic shaker, as used for Sic Bo, can replace the throw.
- **Calls.** The dealer announces the roll, its spread and the Between pay in force before the card, then the card and the winning wagers.
- **Shoe.** Prepare 6 decks of A to 6 (144 cards), shuffle, offer the cut and place the cut card with 36 cards behind it. A standard dealing shoe holds them. With one card a round, a shoe deals 108 rounds.
- **Card counting.** The cards dealt are visible, and Between can be counted at this penetration: the Math Report measures by how much. A live table should shuffle after every round, use a continuous shuffling machine, or place the cut card much nearer the front.
- **The RNG version** draws the dice and the cards from a certified random number generator, deals from a virtual shoe with the same cut card (or reshuffles it every round), and settles at once; the player's tap or throw only animates the dice. It has no physical irregularities. Every rule, limit and pay is the same.

## 10. Irregularities

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- **Die off the table.** A die that leaves the layout (off the table, into the chip rail or the dealer's bank) is no roll: the dealer calls it, checks the dice and the shooter rolls again. Wagers stay as they are.
- **Cocked die.** A die that comes to rest tilted, against a chip, the rail, a card or the other die, so that no face is plainly uppermost, is no roll: the dice are rolled again. A die lying flat on a chip is read as it lies.
- **Dice thrown too early.** Dice thrown before the dealer closes betting are no roll.
- **Misread result.** The dice and cards as they lie decide the round: a misread call is corrected before the round is settled, and a settlement error is corrected before the next round begins.
- **Exposed card.** A card exposed before the roll (dropped, flashed, turned while the shoe is handled) is burned to the discards, and the round proceeds with the next card. The round's card dealt face up instead of face down stands: facing it down is only presentation.
- **Mis-deal.** Only the first card dealt after the roll counts. A card dealt before the roll, or any card after the first, is burned.
- **Cards short of the cut card.** It cannot happen as the shoe is prepared: a round uses at most one card, and 36 cards sit behind the cut card. If the shoe were nonetheless to run out during a round (a cut card placed too deep, cards missing), the dealer shuffles the discards of earlier rounds, never the cards of the round in progress, and completes the round from them; the whole shoe is then shuffled before the next round. This is what the RNG version does.

## To confirm

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These rulings are standard casino practice, not rules the game's code decides. Confirm each one, or replace it with the house's own, before a live table opens.

- Settlement order: losing wagers collected first, then the winners paid, main wager first.
- Dice off the table and cocked dice are no roll, rather than read where they lie.
- At a table with several players, one shooter per round, in turn (or an automatic shaker), whose roll settles every seat.
- No card is burned after the shuffle, as in the RNG version; a burn card would not change the rules.
- An exposed card is burned; the round's card dealt face up in error stands.
- Extra cards, and a card dealt before the roll, are burned.
- Between counting: shuffle after every round or use a continuous shuffler at a live table.

# Dice Spread — Math Report

Will the card land between your dice?

## 1. Cover

| Game      | Dice Spread, an original table game of the Roll & Deal family                                                                    |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|
| Version   | 0.1.0                                                                                                                            |
| Document  | Math Report, revision 3                                                                                                          |
| Date      | 2026-09-30                                                                                                                       |
| Author    | Jonathas Costa                                                                                                                   |
| Code      | commit 47bab22a941c983cd3e18fd6989ebb7a1e6479b5 (2026-09-30): the last change to the engine, its tests or their recorded results |
| Figures   | docs/results.json, written with this report by tools/generate-docs.ts                                                            |
| Companion | Rules of Play ( docs/rules/dice-spread.md )                                                                                      |

### Revision history

| Revision | Date       | Author         | Change                                                                                                                                                                                         |
|----------|------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | 2026-09-30 | Jonathas Costa | First game sheet: rules, declared math, exact and Monte Carlo tests. Card counting exposure measured.                                                                                          |
| 2        | 2026-09-30 | Jonathas Costa | Renamed from its working title, with English names for the game and its bets.                                                                                                                  |
| 3        | 2026-09-30 | Jonathas Costa | Reformatted for submission as two documents, Rules of Play and Math Report, generated from the code with every figure traced to the engine or to a test; card counting measured on every game. |

**How to read this report.** Every figure is read from `docs/results.json`, which the generator builds from the game's `mathSummary()` (the declared figures) and from the figures its tests record (`packages/engine/src/games/dice-spread/results/`); the generator fails if a document holds a number that is not in it. Each table names the test that reproduces it, as file > suite > test: `pnpm test` runs the exact tests and `pnpm test:math` the simulations, and each fails if it no longer reproduces its recorded figures. Returns are per unit staked, the stake included; exact figures are fractions, rounded percentages beside them.

## 2. Summary

| Wager           | RTP      | House edge | Hit frequency | Push frequency | SD per unit | Max payout | Max exposure at max bet |
|-----------------|----------|------------|---------------|----------------|-------------|------------|-------------------------|
| Between (main)  | 96.2963% | 3.7037%    | 18.5185%      | 44.4444%       | 1.1174      | 4×         | 1,000.00 (on 250.00)    |
| Match (side)    | 91.6667% | 8.3333%    | 30.5556%      | —              | 1.3819      | 2×         | 50.00 (on 25.00)        |
| Bullseye (side) | 85.1852% | 14.8148%   | 3.7037%       | —              | 4.3436      | 22×        | 550.00 (on 25.00)       |
| Doubles (side)  | 83.3333% | 16.6667%   | 16.6667%      | —              | 1.8634      | 4×         | 100.00 (on 25.00)       |
| Triple (side)   | 86.1111% | 13.8889%   | 2.7778%       | —              | 5.0944      | 30×        | 750.00 (on 25.00)       |

- **RTP** and **house edge** are per unit staked, pushes included. **Hit frequency** is the chance that the wager wins in a round; **push frequency**, that the stake is simply returned. **SD per unit** is the standard deviation of the net result per unit staked.
- **Max payout** is the largest net win per unit staked, the stake being returned on top. **Max exposure at max bet** is the most a single wager at its maximum can win in a round, found by running the real game over every outcome at the table maximums; all the wagers of one position together can win at most 1,475.00 in a round.

Reproduced by `packages/engine/src/games/dice-spread/dice-spread.test.ts > Dice Spread — the figures of the Math Report > adds up each bet's outcomes to its declared figures, and finds the most a round can pay`. The declared figures are `mathSummary()` (`bets.ts`); the test holds them equal to the enumeration.

## 3. Game model

- **Sample space.**  $\Omega = D \times C$ .  $D$  is the ordered pair of dice ( $d_1, d_2$ ), each uniform on 1 to 6: 36 equally likely rolls.  $C$  is the value  $c$  of the one card, uniform on 1 to 6.  $|\Omega| = 216$  equally likely outcomes; with the card's suit as well, 864, which the exact test also enumerates and which gives the same fractions. Write  $L$  and  $H$  for the lower and higher die and  $S = H - L$  for the spread.
- **Card supply.** The declared figures take the card uniform over its 6 values, independent of the dice: an infinite shoe. That is also exactly what the table's 6-deck shoe returns in the long run to a player who bets the same way every round: each round deals one card and the cut card always comes out after 108 cards, so the rounds a shoe deals never depend on the cards, and over all shuffles the card of any given round is uniform over the values (section six confirms it by simulation).
- **Decisions.** None: every wager is settled by the roll and the card.

## 4. Per-wager analysis

### 4.1 Between (main)

**Wins** when  $S \geq 2$  and  $L < c < H$ , paid by  $S$ ; a push when  $S < 2$ . **Method:** exact enumeration of the production game over  $\Omega$ .

| Outcome                | Probability | Exactly | Pays   | Returns per unit | Contribution to RTP |
|------------------------|-------------|---------|--------|------------------|---------------------|
| Spread 2, card between | 3.7037%     | 1/27    | 4 to 1 | 5                | 5/27                |
| Spread 3, card between | 5.5556%     | 1/18    | 2 to 1 | 3                | 1/6                 |
| Spread 4, card between | 5.5556%     | 1/18    | 1 to 1 | 2                | 1/9                 |
| Spread 5, card between | 3.7037%     | 1/27    | 1 to 2 | 3/2              | 1/18                |
| Spread 1, or a pair    | 44.4444%    | 4/9     | push   | 1                | 4/9                 |
| Any other outcome      | 37.0370%    | 10/27   | loses  | 0                | 0                   |

- **Total return (RTP):** 26/27 (96.2963%), the sum of the contributions.
- **House edge:** 1/27 (3.7037%).
- **Variance** of the net result per unit staked: 3641/2916 (1.2486); **standard deviation** 1.1174.
- **Hit frequency** (the wager wins): 5/27 (18.5185%); **push frequency** 4/9 (44.4444%).

Reproduced by `packages/engine/src/games/dice-spread/dice-spread.test.ts` › *Dice Spread — the figures of the Math Report* › adds up each bet's outcomes to its declared figures, and finds the most a round can pay.

### 4.2 Match (side)

**Wins** when  $c = d_1$  or  $c = d_2$ . **Method:** exact enumeration of the production game over  $\Omega$ .

| Outcome            | Probability | Exactly | Pays   | Returns per unit | Contribution to RTP |
|--------------------|-------------|---------|--------|------------------|---------------------|
| Card matches a die | 30.5556%    | 11/36   | 2 to 1 | 3                | 11/12               |
| Any other outcome  | 69.4444%    | 25/36   | loses  | 0                | 0                   |

- **Total return (RTP):** 11/12 (91.6667%), the sum of the contributions.
- **House edge:** 1/12 (8.3333%).
- **Variance** of the net result per unit staked: 275/144 (1.9097); **standard deviation** 1.3819.
- **Hit frequency** (the wager wins): 11/36 (30.5556%).

Reproduced by `packages/engine/src/games/dice-spread/dice-spread.test.ts` › *Dice Spread — the figures of the Math Report* › adds up each bet's outcomes to its declared figures, and finds the most a round can pay.

### 4.3 Bullseye (side)

**Wins** when  $S = 2$  and  $c$  is the value midway between  $L$  and  $H$ . **Method:** exact enumeration of the production game over  $\Omega$ .

| Outcome                            | Probability | Exactly | Pays    | Returns per unit | Contribution to RTP |
|------------------------------------|-------------|---------|---------|------------------|---------------------|
| Spread 2, card is the middle value | 3.7037%     | 1/27    | 22 to 1 | 23               | 23/27               |
| Any other outcome                  | 96.2963%    | 26/27   | loses   | 0                | 0                   |

- **Total return (RTP):** 23/27 (85.1852%), the sum of the contributions.
- **House edge:** 4/27 (14.8148%).
- **Variance** of the net result per unit staked: 13754/729 (18.8669); **standard deviation** 4.3436.
- **Hit frequency** (the wager wins): 1/27 (3.7037%).

Reproduced by `packages/engine/src/games/dice-spread/dice-spread.test.ts` › *Dice Spread — the figures of the Math Report* › adds up each bet's outcomes to its declared figures, and finds the most a round can pay.

### 4.4 Doubles (side)

**Wins** when  $d_1 = d_2$ . **Method:** exact enumeration of the production game over  $\Omega$ .

| Outcome             | Probability | Exactly | Pays   | Returns per unit | Contribution to RTP |
|---------------------|-------------|---------|--------|------------------|---------------------|
| The dice are a pair | 16.6667%    | 1/6     | 4 to 1 | 5                | 5/6                 |
| Any other outcome   | 83.3333%    | 5/6     | loses  | 0                | 0                   |

- **Total return (RTP):** 5/6 (83.3333%), the sum of the contributions.
- **House edge:** 1/6 (16.6667%).
- **Variance** of the net result per unit staked: 125/36 (3.4722); **standard deviation** 1.8634.
- **Hit frequency** (the wager wins): 1/6 (16.6667%).

Reproduced by `packages/engine/src/games/dice-spread/dice-spread.test.ts` › *Dice Spread — the figures of the Math Report* › adds up each bet's outcomes to its declared figures, and finds the most a round can pay.

### 4.5 Triple (side)

**Wins** when  $d_1 = d_2 = c$ . **Method:** exact enumeration of the production game over  $\Omega$ .

| Outcome                         | Probability | Exactly | Pays    | Returns per unit | Contribution to RTP |
|---------------------------------|-------------|---------|---------|------------------|---------------------|
| A pair, and the card matches it | 2.7778%     | 1/36    | 30 to 1 | 31               | 31/36               |
| Any other outcome               | 97.2222%    | 35/36   | loses   | 0                | 0                   |

- **Total return (RTP):** 31/36 (86.1111%), the sum of the contributions.
- **House edge:** 5/36 (13.8889%).
- **Variance** of the net result per unit staked: 33635/1296 (25.9529); **standard deviation** 5.0944.
- **Hit frequency** (the wager wins): 1/36 (2.7778%).

Reproduced by `packages/engine/src/games/dice-spread/dice-spread.test.ts` › *Dice Spread — the figures of the Math Report* › adds up each bet's outcomes to its declared figures, and finds the most a round can pay.

## 5. Progressive analysis

Not applicable: Dice Spread has no progressive wager.

## 6. Finite-shoe effects

As section three shows, the 6-deck shoe at 75% penetration returns exactly the declared figures to a flat bettor: there is no finite-shoe shift to measure, only its absence to confirm. The simulation of section seven deals from that shoe, with its cut card and reshuffles:

| Wager    | RTP, declared | RTP, six-deck shoe (observed) | Difference | Standard error |
|----------|---------------|-------------------------------|------------|----------------|
| Between  | 96.296%       | 96.289%                       | −0.007 pp  | 0.010 pp       |
| Match    | 91.667%       | 91.676%                       | +0.009 pp  | 0.012 pp       |
| Bullseye | 85.185%       | 85.205%                       | +0.020 pp  | 0.039 pp       |
| Doubles  | 83.333%       | 83.376%                       | +0.043 pp  | 0.017 pp       |
| Triple   | 86.111%       | 86.125%                       | +0.014 pp  | 0.046 pp       |

Every difference lies within the run's noise. *Reproduced by* `packages/engine/src/games/dice-spread/dice-spread.math.test.ts` > *Dice Spread — Monte Carlo against the real six-deck shoe > lands every bet within ±0.15 pp of its declared RTP.*

## 7. Simulation verification

| Rounds    | 124,852,059                                                                                                                                                                                                     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Generator | xoshiro128** 1.1 (Blackman & Vigna), its state expanded from the seed by SplitMix64; a string seed is first hashed with FNV-1a (64-bit) over its UTF-8 bytes                                                    |
| Seed      | dice-spread/monte-carlo                                                                                                                                                                                         |
| Cards     | the 6-deck shoe of 144 cards, ranks 1 to 6, with its cut card after 108 cards (penetration 75%) and reshuffles, as the production game deals it                                                                 |
| Stakes    | every wager at its minimum each round: Between 0.50, Match 0.50, Bullseye 0.50, Doubles 0.50 and Triple 0.50                                                                                                    |
| Criterion | each RTP within ±0.15 pp of the declared figure, and each hit and push frequency within 3.29 binomial standard errors; the round count makes ±0.15 pp equal to 3.29 standard errors for the most volatile wager |

| Figure                  | Observed | Expected | Difference | Standard error | 95% confidence interval | Allowed   | Result |
|-------------------------|----------|----------|------------|----------------|-------------------------|-----------|--------|
| Between, RTP            | 96.289%  | 96.296%  | −0.007 pp  | 0.010 pp       | 96.270% to 96.309%      | ±0.150 pp | pass   |
| Between, hit frequency  | 18.515%  | 18.519%  | −0.004 pp  | 0.003 pp       | 18.508% to 18.522%      | ±0.011 pp | pass   |
| Between, push frequency | 44.446%  | 44.444%  | +0.002 pp  | 0.004 pp       | 44.437% to 44.455%      | ±0.015 pp | pass   |
| Match, RTP              | 91.676%  | 91.667%  | +0.009 pp  | 0.012 pp       | 91.652% to 91.700%      | ±0.150 pp | pass   |
| Match, hit frequency    | 30.559%  | 30.556%  | +0.003 pp  | 0.004 pp       | 30.551% to 30.567%      | ±0.014 pp | pass   |
| Bullseye, RTP           | 85.205%  | 85.185%  | +0.020 pp  | 0.039 pp       | 85.129% to 85.281%      | ±0.150 pp | pass   |
| Bullseye, hit frequency | 3.705%   | 3.704%   | +0.001 pp  | 0.002 pp       | 3.701% to 3.708%        | ±0.006 pp | pass   |
| Doubles, RTP            | 83.376%  | 83.333%  | +0.043 pp  | 0.017 pp       | 83.344% to 83.409%      | ±0.150 pp | pass   |
| Doubles, hit frequency  | 16.675%  | 16.667%  | +0.009 pp  | 0.003 pp       | 16.669% to 16.682%      | ±0.011 pp | pass   |
| Triple, RTP             | 86.125%  | 86.111%  | +0.014 pp  | 0.046 pp       | 86.035% to 86.214%      | ±0.150 pp | pass   |
| Triple, hit frequency   | 2.778%   | 2.778%   | 0.000 pp   | 0.001 pp       | 2.775% to 2.781%        | ±0.005 pp | pass   |

The 95% confidence interval is the observed value ± 1.96 standard errors, and at that level about one figure in twenty falls outside its interval by chance. The tests' own criterion, "Allowed" above, is wider: 3.29 standard errors, or a fixed tolerance. *Reproduced by* `packages/engine/src/games/dice-spread/dice-spread.math.test.ts` > *Dice Spread — Monte Carlo against the real six-deck shoe > lands every bet within ±0.15 pp of its declared RTP.*

## 8. Assumptions and limitations

**Independence.** The dice and the cards are independent: the RNG draws each die and each card separately (a live table uses separate devices), and nothing about the dice changes the shoe. Within a round the cards are dealt from one shoe, which the finite-shoe figures account for.

**Card counting.** A player who tracks the cards dealt knows the shoe's composition before every round. Between's value depends on it: over the 36 rolls, one unit is worth, by the card's value:

| Card value | Between, expected net | Bullseye, expected net |
|------------|-----------------------|------------------------|
| 1          | −0.5556               | −1.0000                |
| 2          | +0.0833               | +0.2778                |
| 3          | +0.3611               | +0.2778                |
| 4          | +0.3611               | +0.2778                |
| 5          | +0.0833               | +0.2778                |
| 6          | −0.5556               | −1.0000                |

The other wagers are worth the same whatever the card, so they cannot be counted. The exposure, computed exactly for a player who knows the composition before every round (a hypergeometric model of the shoe, no simulation):

| Penetration     | Rounds per shoe | Between: rounds favouring the counter | Counter's edge in them | Break-even bet spread | Bullseye: rounds favouring the counter |
|-----------------|-----------------|---------------------------------------|------------------------|-----------------------|----------------------------------------|
| 75% (the table) | 108             | 8.57%                                 | 2.05%                  | 1 to 22.1             | 0.47%                                  |
| 50%             | 72              | 3.66%                                 | 1.18%                  | 1 to 86.7             | 0.01%                                  |
| 25%             | 36              | 0.47%                                 | 0.59%                  | 1 to 1,329.0          | 0.00%                                  |

The table's limits allow a spread of 1 to 500 on Between, so at the table's penetration a skilled counter could beat it wherever the cards are visible. A live table should shuffle after every round, use a continuous shuffling machine, or cut the penetration well below the table's; an RNG table that reshuffles every round returns exactly the declared figures. *Reproduced by* `packages/engine/src/games/dice-spread/counting.test.ts` > *Dice Spread — card counting exposure > measures the countable bets at 75%, 50% and 25% penetration, against the spread the limits allow.*

**Rounding.** Winnings are paid to the cent, rounded down (the engine's `winnings()`); the declared figures assume exact pays. Every odds in these games has a denominator of one or two, so a wager in multiples of the minimum pays exactly and the rounding never applies at the table's stakes.

**No player or dealer error** is modelled, nor any irregularity: the figures are those of the rules as the engine settles them.

**Randomness.** The analysis assumes a certified RNG: every die face and every card position equally likely and independent. The engine draws integers by rejection sampling, without modulo bias, and shuffles by Fisher-Yates in a specified order.

## 9. Appendix

**Full enumeration.** Every distinct roll (the higher die first; "ways" counts the ordered rolls that show it) against every card value, with its probability and each wager's net result per unit staked (Between's push is 0). Each column, weighted by the probabilities, adds up to the wager's RTP; the test checks it.

| Dice | Ways | Card | Probability | Between | Match | Bullseye | Doubles | Triple |
|------|------|------|-------------|---------|-------|----------|---------|--------|
| 1-1  | 1    | 1    | 1/216       | 0       | 2     | -1       | 4       | 30     |
| 1-1  | 1    | 2    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 1-1  | 1    | 3    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 1-1  | 1    | 4    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 1-1  | 1    | 5    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 1-1  | 1    | 6    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 2-1  | 2    | 1    | 1/108       | 0       | 2     | -1       | -1      | -1     |
| 2-1  | 2    | 2    | 1/108       | 0       | 2     | -1       | -1      | -1     |
| 2-1  | 2    | 3    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 2-1  | 2    | 4    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 2-1  | 2    | 5    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 2-1  | 2    | 6    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 3-1  | 2    | 1    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 3-1  | 2    | 2    | 1/108       | 4       | -1    | 22       | -1      | -1     |
| 3-1  | 2    | 3    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 3-1  | 2    | 4    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 3-1  | 2    | 5    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 3-1  | 2    | 6    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 4-1  | 2    | 1    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 4-1  | 2    | 2    | 1/108       | 2       | -1    | -1       | -1      | -1     |
| 4-1  | 2    | 3    | 1/108       | 2       | -1    | -1       | -1      | -1     |
| 4-1  | 2    | 4    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 4-1  | 2    | 5    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 4-1  | 2    | 6    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 5-1  | 2    | 1    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 5-1  | 2    | 2    | 1/108       | 1       | -1    | -1       | -1      | -1     |
| 5-1  | 2    | 3    | 1/108       | 1       | -1    | -1       | -1      | -1     |
| 5-1  | 2    | 4    | 1/108       | 1       | -1    | -1       | -1      | -1     |
| 5-1  | 2    | 5    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 5-1  | 2    | 6    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 6-1  | 2    | 1    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 6-1  | 2    | 2    | 1/108       | 1/2     | -1    | -1       | -1      | -1     |
| 6-1  | 2    | 3    | 1/108       | 1/2     | -1    | -1       | -1      | -1     |
| 6-1  | 2    | 4    | 1/108       | 1/2     | -1    | -1       | -1      | -1     |
| 6-1  | 2    | 5    | 1/108       | 1/2     | -1    | -1       | -1      | -1     |
| 6-1  | 2    | 6    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 2-2  | 1    | 1    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 2-2  | 1    | 2    | 1/216       | 0       | 2     | -1       | 4       | 30     |
| 2-2  | 1    | 3    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 2-2  | 1    | 4    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 2-2  | 1    | 5    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 2-2  | 1    | 6    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 3-2  | 2    | 1    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 3-2  | 2    | 2    | 1/108       | 0       | 2     | -1       | -1      | -1     |
| 3-2  | 2    | 3    | 1/108       | 0       | 2     | -1       | -1      | -1     |
| 3-2  | 2    | 4    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 3-2  | 2    | 5    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 3-2  | 2    | 6    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 4-2  | 2    | 1    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 4-2  | 2    | 2    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 4-2  | 2    | 3    | 1/108       | 4       | -1    | 22       | -1      | -1     |
| 4-2  | 2    | 4    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 4-2  | 2    | 5    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 4-2  | 2    | 6    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 5-2  | 2    | 1    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 5-2  | 2    | 2    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 5-2  | 2    | 3    | 1/108       | 2       | -1    | -1       | -1      | -1     |
| 5-2  | 2    | 4    | 1/108       | 2       | -1    | -1       | -1      | -1     |
| 5-2  | 2    | 5    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 5-2  | 2    | 6    | 1/108       | -1      | -1    | -1       | -1      | -1     |

| Dice | Ways | Card | Probability | Between | Match | Bullseye | Doubles | Triple |
|------|------|------|-------------|---------|-------|----------|---------|--------|
| 6-2  | 2    | 1    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 6-2  | 2    | 2    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 6-2  | 2    | 3    | 1/108       | 1       | -1    | -1       | -1      | -1     |
| 6-2  | 2    | 4    | 1/108       | 1       | -1    | -1       | -1      | -1     |
| 6-2  | 2    | 5    | 1/108       | 1       | -1    | -1       | -1      | -1     |
| 6-2  | 2    | 6    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 3-3  | 1    | 1    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 3-3  | 1    | 2    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 3-3  | 1    | 3    | 1/216       | 0       | 2     | -1       | 4       | 30     |
| 3-3  | 1    | 4    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 3-3  | 1    | 5    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 3-3  | 1    | 6    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 4-3  | 2    | 1    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 4-3  | 2    | 2    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 4-3  | 2    | 3    | 1/108       | 0       | 2     | -1       | -1      | -1     |
| 4-3  | 2    | 4    | 1/108       | 0       | 2     | -1       | -1      | -1     |
| 4-3  | 2    | 5    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 4-3  | 2    | 6    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 5-3  | 2    | 1    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 5-3  | 2    | 2    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 5-3  | 2    | 3    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 5-3  | 2    | 4    | 1/108       | 4       | -1    | 22       | -1      | -1     |
| 5-3  | 2    | 5    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 5-3  | 2    | 6    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 6-3  | 2    | 1    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 6-3  | 2    | 2    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 6-3  | 2    | 3    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 6-3  | 2    | 4    | 1/108       | 2       | -1    | -1       | -1      | -1     |
| 6-3  | 2    | 5    | 1/108       | 2       | -1    | -1       | -1      | -1     |
| 6-3  | 2    | 6    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 4-4  | 1    | 1    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 4-4  | 1    | 2    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 4-4  | 1    | 3    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 4-4  | 1    | 4    | 1/216       | 0       | 2     | -1       | 4       | 30     |
| 4-4  | 1    | 5    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 4-4  | 1    | 6    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 5-4  | 2    | 1    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 5-4  | 2    | 2    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 5-4  | 2    | 3    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 5-4  | 2    | 4    | 1/108       | 0       | 2     | -1       | -1      | -1     |
| 5-4  | 2    | 5    | 1/108       | 0       | 2     | -1       | -1      | -1     |
| 5-4  | 2    | 6    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 6-4  | 2    | 1    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 6-4  | 2    | 2    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 6-4  | 2    | 3    | 1/108       | -1      | -1    | -1       | -1      | -1     |
| 6-4  | 2    | 4    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 6-4  | 2    | 5    | 1/108       | 4       | -1    | 22       | -1      | -1     |
| 6-4  | 2    | 6    | 1/108       | -1      | 2     | -1       | -1      | -1     |
| 5-5  | 1    | 1    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 5-5  | 1    | 2    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 5-5  | 1    | 3    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 5-5  | 1    | 4    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 5-5  | 1    | 5    | 1/216       | 0       | 2     | -1       | 4       | 30     |
| 5-5  | 1    | 6    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 6-5  | 2    | 1    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 6-5  | 2    | 2    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 6-5  | 2    | 3    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 6-5  | 2    | 4    | 1/108       | 0       | -1    | -1       | -1      | -1     |
| 6-5  | 2    | 5    | 1/108       | 0       | 2     | -1       | -1      | -1     |
| 6-5  | 2    | 6    | 1/108       | 0       | 2     | -1       | -1      | -1     |
| 6-6  | 1    | 1    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 6-6  | 1    | 2    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 6-6  | 1    | 3    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 6-6  | 1    | 4    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 6-6  | 1    | 5    | 1/216       | 0       | -1    | -1       | 4       | -1     |
| 6-6  | 1    | 6    | 1/216       | 0       | 2     | -1       | 4       | 30     |

Reproduced by `packages/engine/src/games/dice-spread/dice-spread.test.ts` › *Dice Spread — the figures of the Math Report* › *settles every roll and card as the rules state: the enumeration adds up to every RTP.*

**Records.** The tests recorded these figures ( `packages/engine/src/games/dice-spread/results/` ):

| Record                        | Test                                                                                                                                                                                                            |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>counting.json</code>    | <code>packages/engine/src/games/dice-spread/counting.test.ts</code> › Dice Spread — card counting exposure › measures the countable bets at 75%, 50% and 25% penetration, against the spread the limits allow   |
| <code>enumeration.json</code> | <code>packages/engine/src/games/dice-spread/dice-spread.test.ts</code> › Dice Spread — the figures of the Math Report › settles every roll and card as the rules state: the enumeration adds up to every RTP    |
| <code>exact.json</code>       | <code>packages/engine/src/games/dice-spread/dice-spread.test.ts</code> › Dice Spread — the figures of the Math Report › adds up each bet's outcomes to its declared figures, and finds the most a round can pay |
| <code>monte-carlo.json</code> | <code>packages/engine/src/games/dice-spread/dice-spread.math.test.ts</code> › Dice Spread — Monte Carlo against the real six-deck shoe › lands every bet within $\pm 0.15$ pp of its declared RTP               |

# Moving Target — Rules of Play

Your dice set the target. Will the cards land on it?

## 1. Game and document

| Game       | Moving Target, an original table game of the Roll & Deal family                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Version    | 0.1.0                                                                                                                             |
| Document   | Rules of Play, revision 3                                                                                                         |
| Date       | 2026-09-30                                                                                                                        |
| Author     | Jonathas Costa                                                                                                                    |
| Companion  | Math Report ( docs/math/moving-target.md )                                                                                        |
| Demo table | <a href="https://jonathascosta.github.io/CasinoGames/moving-target">https://jonathascosta.github.io/CasinoGames/moving-target</a> |

### Revision history

| Revision | Date       | Author         | Change                                                                                                                                                                                         |
|----------|------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | 2026-09-30 | Jonathas Costa | First game sheet: rules, declared math, exact and Monte Carlo tests. The six-deck shoe's shift measured.                                                                                       |
| 2        | 2026-09-30 | Jonathas Costa | Renamed from its working title, with English names for the game and its bets.                                                                                                                  |
| 3        | 2026-09-30 | Jonathas Costa | Reformatted for submission as two documents, Rules of Play and Math Report, generated from the code with every figure traced to the engine or to a test; card counting measured on every game. |

Money is written in the table's currency with two decimals, as the demo shows it (the demo plays with virtual chips).

## 2. Objective

The player rolls two dice, whose sum sets the round's target, and the dealer deals cards one at a time, adding their values, until the total reaches or passes the target. The main wager, Exact Hit, wins when the total lands exactly on the target and pays more on the targets that are harder to hit; optional side wagers settle on the same deal.

## 3. Equipment

- **Dice.** Two standard six-sided dice, faces 1 to 6, thrown by the player (by the round's shooter at a live table: see section nine). Their **sum**, from 2 to 12, is the round's **target**.
- **Shoe.** 6 decks, each of the A to the 10 in 4 suits (40 cards): 240 cards in all, made from standard decks with every card above the 10 removed: no jacks, queens or kings. The ace counts 1 and every other card its number, up to 10.
- **Cut card.** It goes in with 60 cards behind it, a penetration of 75%: 180 cards are dealt before it comes out. When it comes out, the round in progress is completed, and the whole shoe is shuffled before the next round.
- **Cosmetic.** Suits never count, only values. Which die is which does not matter: only their sum does. The order of the cards does matter, as it decides when the deal stops.
- **Also.** The layout (section four), chips, a discard rack and, at a live table, a dice cup or shaker and a marker for the target.

## 4. Table layout

Each player position has one spot for the main wager, Exact Hit, in the centre, and one on each side for First Card and 3+ Cards. The felt prints Exact Hit's pays by target and the side wagers' pays:

| Target | Exact Hit pays | A 1.00 wager wins |
|--------|----------------|-------------------|
| 2      | 15 to 2        | 7.50              |
| 3      | 7 to 1         | 7.00              |
| 4      | 6 to 1         | 6.00              |
| 5      | 11 to 2        | 5.50              |
| 6      | 5 to 1         | 5.00              |
| 7      | 9 to 2         | 4.50              |
| 8      | 4 to 1         | 4.00              |
| 9      | 7 to 2         | 3.50              |
| 10     | 3 to 1         | 3.00              |
| 11     | 5 to 1         | 5.00              |
| 12     | 5 to 1         | 5.00              |

| Side wager | Wins when                          | Pays   | A 1.00 wager wins |
|------------|------------------------------------|--------|-------------------|
| First Card | the first card alone is the target | 9 to 1 | 9.00              |
| 3+ Cards   | the dealer needs 3 cards or more   | 4 to 1 | 4.00              |

The demo shows this layout at <https://jonathascosta.github.io/CasinoGames/moving-target>. At the top are the dice and the target board: the target, locked in large after the roll with what Exact Hit pays on it, Exact Hit's payable with that target lit, the dealer's running total filling up to the target, and the count of cards that 3+ Cards watches. The dealer's cards land beside it, one at a time. Below is the felt, with Exact Hit in the centre and a side wager on each side, then the chips, the balance, the total wagered and the Clear, Roll and Autoplay buttons, with Rules and Paytable buttons beside the balance; an RTP monitor below the table compares the rounds played with the declared figures.

## 5. Wagers

---

Every wager is placed before the roll and settled once the dealer stops. The main wager comes first; the side wagers follow in the felt's order. The game has no progressive wager.

### 5.1 Exact Hit (main wager)

- **When.** Before the roll. It can be placed alone, and every side wager needs it.
- **Limits.** 0.50 to 250.00.
- **Wins** when the dealer's total lands exactly on the target.
- **Pays** by the target, as the felt prints it: 15 to 2 on 2, 7 to 1 on 3, 6 to 1 on 4, 11 to 2 on 5, 5 to 1 on 6, 9 to 2 on 7, 4 to 1 on 8, 7 to 2 on 9, 3 to 1 on 10, 5 to 1 on 11 and 5 to 1 on 12. Half-unit odds pay exactly on the table's chips: a 1.00 wager on a target of 2 wins 7.50.
- **Pushes** never.
- **Loses** when the total passes the target.

### 5.2 First Card (side wager)

- **When.** Before the roll, only with an Exact Hit wager. The target is not known when it is placed.
- **Limits.** 0.50 to 25.00.
- **Wins** when the first card alone is the target. No card is worth more than 10, so it cannot win on a target above 10.
- **Pays** 9 to 1.
- **Pushes** never.
- **Loses** on any other deal.

### 5.3 3+ Cards (side wager)

- **When.** Before the roll, only with an Exact Hit wager.
- **Limits.** 0.50 to 25.00.
- **Wins** when the dealer needs 3 cards or more to reach or pass the target, whether the total then lands on it or passes it.
- **Pays** 4 to 1.
- **Pushes** never.
- **Loses** when one or two cards reach the target.

## 6. Sequence of play

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1. **Wagers.** The dealer invites wagers. Each player places Exact Hit and any side wagers, within the limits. The dealer then closes betting ("No more bets").
2. **Shuffle, if due.** If the cut card came out during the previous round, the dealer shuffles the whole shoe, has it cut and places the cut card with 60 cards behind it before the round begins.
3. **Roll.** The player rolls both dice. The dealer announces their sum as the target, with what Exact Hit pays on it, and marks the target on the layout.
4. **Deal.** The dealer deals cards from the shoe face up, one at a time, announcing each card and the running total. The dealer stops as soon as the total equals or passes the target, and never deals a card once it has.
5. **Compare.** The dealer compares the final total with the target (section seven) and counts the cards dealt.
6. **Settle.** The dealer settles every wager (section eight), clears the cards to the discard rack and returns the dice for the next round.

## 7. Comparing the total with the target

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There are no hands to rank: the dealer's total is compared with the target.

- The **total** is the sum of the cards dealt so far, the ace counting 1.
- The total **hits** when it equals the target, and goes **over** when it passes it; either way the deal stops.
- The **card count** is the number of cards the dealer needed to stop, from 1 to 12 (a target of 12 reached by aces alone).

Worked examples:

- Dice 4-3 (target 7), cards 3 and 4 (running total 3, 7): the total lands on the target. Exact Hit wins 9 to 2; First Card and 3+ Cards lose.
- Dice 4-1 (target 5), one card, a 5: the total lands on the target. Exact Hit wins 11 to 2; First Card wins 9 to 1; 3+ Cards loses.
- Dice 6-6 (target 12), cards 10 and 5 (running total 10, 15): the total passes the target. Exact Hit, First Card and 3+ Cards lose.
- Dice 6-5 (target 11), cards 1, 2, 3 and 7 (running total 1, 3, 6, 13): the total passes the target. 3+ Cards wins 4 to 1; Exact Hit and First Card lose.

## 8. Settlement procedure

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- The dealer settles once the deal stops: first collecting every losing wager, then paying the winners, Exact Hit first and then First Card and 3+ Cards.
- A winning wager is paid at its odds and keeps its stake. The tables above show what a 1.00 wager wins on each line.
- A pay that is not a whole number of cents (half-unit odds on an odd number of cents) is rounded down to the cent.
- There is no fee, commission or progressive meter.

## 9. Physical and live-dealer adaptation

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- **Dice.** The player throws from a dice cup or by hand, across the layout, so that both dice tumble and come to rest on the felt. At a table with several players, one shooter throws per round, in turn, and that one roll and one deal settle every seat; an automatic shaker, as used for Sic Bo, can replace the throw.
- **Calls and markers.** The dealer marks the target on the layout (a puck on a printed row of targets) and announces the running total after every card, then "hit" or "over" and the card count.
- **Shoe.** Prepare 6 decks of A to 10 (240 cards: every jack, queen and king removed), shuffle, offer the cut and place the cut card with 60 cards behind it. A round uses one to 12 cards, so the number of rounds a shoe deals varies.
- **Card counting.** The cards are dealt face up, and Exact Hit and 3+ Cards can be counted at this penetration: the Math Report measures by how much. A live table should use a continuous shuffling machine, or shuffle much earlier.
- **The RNG version** draws the dice and the cards from a certified random number generator, deals from a virtual shoe with the same cut card (or reshuffles it every round), and settles at once; the player's tap or throw only animates the dice, and the demo lands each card face down and turns it after a short pause, which is only presentation. It has no physical irregularities. Every rule, limit and pay is the same.

## 10. Irregularities

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- **Die off the table.** A die that leaves the layout (off the table, into the chip rail or the dealer's bank) is no roll: the dealer calls it, checks the dice and the shooter rolls again. Wagers stay as they are.
- **Cocked die.** A die that comes to rest tilted, against a chip, the rail, a card or the other die, so that no face is plainly uppermost, is no roll: the dice are rolled again. A die lying flat on a chip is read as it lies.
- **Dice thrown too early.** Dice thrown before the dealer closes betting are no roll.
- **Misread result.** The dice and cards as they lie decide the round: a misread call is corrected before the round is settled, and a settlement error is corrected before the next round begins.
- **Exposed card.** A card exposed out of turn (dropped, flashed, turned while the shoe is handled) is burned to the discards, and dealing continues with the next card.
- **Mis-deal.** A card dealt after the total reached the target does not count and is burned. A deal stopped too early continues until the total reaches the target; the round is not settled before it does. A card dealt before the roll is burned.
- **Cards short of the cut card.** It cannot happen as the shoe is prepared: a round uses at most 12 cards, and 60 cards sit behind the cut card. If the shoe were nonetheless to run out during a round (a cut card placed too deep, cards missing), the dealer shuffles the discards of earlier rounds, never the cards of the round in progress, and completes the round from them; the whole shoe is then shuffled before the next round. This is what the RNG version does.

## To confirm

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These rulings are standard casino practice, not rules the game's code decides. Confirm each one, or replace it with the house's own, before a live table opens.

- Settlement order: losing wagers collected first, then the winners paid, main wager first.
- Dice off the table and cocked dice are no roll, rather than read where they lie.
- At a table with several players, one shooter per round, in turn (or an automatic shaker), whose roll settles every seat.
- No card is burned after the shuffle, as in the RNG version; a burn card would not change the rules.
- An exposed card is burned and dealing continues with the next card; cards dealt past the target are burned.
- Counting: a continuous shuffler, or a much earlier shuffle, at a live table.

# Moving Target — Math Report

Your dice set the target. Will the cards land on it?

## 1. Cover

| Game      | Moving Target, an original table game of the Roll & Deal family                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|
| Version   | 0.1.0                                                                                                                            |
| Document  | Math Report, revision 3                                                                                                          |
| Date      | 2026-09-30                                                                                                                       |
| Author    | Jonathas Costa                                                                                                                   |
| Code      | commit 47bab22a941c983cd3e18fd6989ebb7a1e6479b5 (2026-09-30): the last change to the engine, its tests or their recorded results |
| Figures   | docs/results.json, written with this report by tools/generate-docs.ts                                                            |
| Companion | Rules of Play ( docs/rules/moving-target.md )                                                                                    |

### Revision history

| Revision | Date       | Author         | Change                                                                                                                                                                                         |
|----------|------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | 2026-09-30 | Jonathas Costa | First game sheet: rules, declared math, exact and Monte Carlo tests. The six-deck shoe's shift measured.                                                                                       |
| 2        | 2026-09-30 | Jonathas Costa | Renamed from its working title, with English names for the game and its bets.                                                                                                                  |
| 3        | 2026-09-30 | Jonathas Costa | Reformatted for submission as two documents, Rules of Play and Math Report, generated from the code with every figure traced to the engine or to a test; card counting measured on every game. |

**How to read this report.** Every figure is read from `docs/results.json`, which the generator builds from the game's `mathSummary()` (the declared figures) and from the figures its tests record (`packages/engine/src/games/moving-target/results/`); the generator fails if a document holds a number that is not in it. Each table names the test that reproduces it, as file > suite > test: `pnpm test` runs the exact tests and `pnpm test:math` the simulations, and each fails if it no longer reproduces its recorded figures. Returns are per unit staked, the stake included; exact figures are fractions, rounded percentages beside them.

## 2. Summary

| Wager             | RTP      | House edge | Hit frequency | SD per unit | Max payout | Max exposure at max bet |
|-------------------|----------|------------|---------------|-------------|------------|-------------------------|
| Exact Hit (main)  | 96.1516% | 3.8484%    | 17.3010%      | 2.1478      | 7.5×       | 1,875.00 (on 250.00)    |
| First Card (side) | 91.6667% | 8.3333%    | 9.1667%       | 2.8855      | 9×         | 225.00 (on 25.00)       |
| 3+ Cards (side)   | 89.5833% | 10.4167%   | 17.9167%      | 1.9175      | 4×         | 100.00 (on 25.00)       |

- **RTP** and **house edge** are per unit staked, pushes included. **Hit frequency** is the chance that the wager wins in a round. **SD per unit** is the standard deviation of the net result per unit staked.
- **Max payout** is the largest net win per unit staked, the stake being returned on top. **Max exposure at max bet** is the most a single wager at its maximum can win in a round, found by running the real game over every outcome at the table maximums; all the wagers of one position together can win at most 2,075.00 in a round.

Reproduced by `packages/engine/src/games/moving-target/moving-target.test.ts` > *Moving Target — exact math* > *the real game over every roll and card sequence* > *breaks every bet down by target, adding up to its declared figures, and finds the most a round can pay*. The declared figures are `mathSummary()` (`bets.ts`); the test holds them equal to the enumeration.

## 3. Game model

- **Sample space.** The dice ( $d_1$ ,  $d_2$ ) are uniform on 1 to 6: 36 rolls, whose sum is the target  $T$ , from 2 to 12. The cards  $c_1$ ,  $c_2$ , ... take values 1 to 10, and the deal stops at the first  $K$  with  $c_1 + \dots + c_K \geq T$ .  $\Omega$  is the set of rolls and the card sequences each can deal: 71,469 outcomes (from 19 sequences for a target of 2 to 18,424 for 12), each with the product of its draws' chances.
- **Card supply.** The declared figures take every card value equally likely and independent: an infinite shoe, as an RNG game that draws each card independently deals it. With every value equally likely, the running total, which starts at zero, passes through  $n$  with chance  $h(n) = (h(n - 1) + \dots + h(n - 10)) / 10$ , and Exact Hit on a target  $T$  wins with chance  $h(T)$ ; `bets.ts` declares these chances in closed form, and the exact test checks the closed forms against the recursion and the game. The real 6-deck shoe deals without replacement and its rounds interact; section six measures by how much that moves each figure.
- **Decisions.** None: every wager is settled by the roll and the deal.

## 4. Per-wager analysis

### 4.1 Exact Hit (main)

**Wins** when the running total  $X_k = c_1 + \dots + c_k$  takes the value  $T$  for some  $k$ , that is  $X_K = T$  where  $K$  is the first  $k$  with  $X_k \geq T$ ; paid at the odds of  $T$ . **Method:** a memoised recursion over the running total and the card count, written from the rules alone, checked against the closed forms of `bets.ts` and against exact enumeration of the production game over  $\Omega$ ; all three agree as fractions.

By target, weighted by the dice:

| Target | P(target) | P(win   target)       | Pays    | Return given target | House edge given target | P(target and win) | Contribution to RTP |
|--------|-----------|-----------------------|---------|---------------------|-------------------------|-------------------|---------------------|
| 2      | 1/36      | 11/100 (11.00%)       | 15 to 2 | 93.500%             | 6.500%                  | 0.3056%           | 2.5972%             |
| 3      | 1/18      | 121/1000 (12.10%)     | 7 to 1  | 96.800%             | 3.200%                  | 0.6722%           | 5.3778%             |
| 4      | 1/12      | 1331/10000 (13.31%)   | 6 to 1  | 93.170%             | 6.830%                  | 1.1092%           | 7.7642%             |
| 5      | 1/9       | 14641/100000 (14.64%) | 11 to 2 | 95.166%             | 4.833%                  | 1.6268%           | 10.5741%            |

| Target      | P(target) | P(win   target)                     | Pays   | Return given target | House edge given target | P(target and win) | Contribution to RTP |
|-------------|-----------|-------------------------------------|--------|---------------------|-------------------------|-------------------|---------------------|
| 6           | 5/36      | 161051/1000000 (16.11%)             | 5 to 1 | 96.631%             | 3.369%                  | 2.2368%           | 13.4209%            |
| 7           | 1/6       | 1771561/10000000 (17.72%)           | 9 to 2 | 97.436%             | 2.564%                  | 2.9526%           | 16.2393%            |
| 8           | 5/36      | 19487171/100000000 (19.49%)         | 4 to 1 | 97.436%             | 2.564%                  | 2.7066%           | 13.5328%            |
| 9           | 1/9       | 214358881/1000000000 (21.44%)       | 7 to 2 | 96.461%             | 3.539%                  | 2.3818%           | 10.7179%            |
| 10          | 1/12      | 2357947691/10000000000 (23.58%)     | 3 to 1 | 94.318%             | 5.682%                  | 1.9650%           | 7.8598%             |
| 11          | 1/18      | 15937424601/100000000000 (15.94%)   | 5 to 1 | 95.625%             | 4.375%                  | 0.8854%           | 5.3125%             |
| 12          | 1/36      | 165311670611/1000000000000 (16.53%) | 5 to 1 | 99.187%             | 0.813%                  | 0.4592%           | 2.7552%             |
| All targets |           |                                     |        |                     |                         |                   | 96.1516%            |

As an outcome table:

| Outcome           | Probability | Exactly                       | Pays    | Returns per unit | Contribution to RTP        |
|-------------------|-------------|-------------------------------|---------|------------------|----------------------------|
| Total exactly 2   | 0.3056%     | 11/3600                       | 15 to 2 | 17/2             | 187/7200                   |
| Total exactly 3   | 0.6722%     | 121/18000                     | 7 to 1  | 8                | 121/2250                   |
| Total exactly 4   | 1.1092%     | 1331/120000                   | 6 to 1  | 7                | 9317/120000                |
| Total exactly 5   | 1.6268%     | 14641/900000                  | 11 to 2 | 13/2             | 190333/1800000             |
| Total exactly 6   | 2.2368%     | 161051/7200000                | 5 to 1  | 6                | 161051/1200000             |
| Total exactly 7   | 2.9526%     | 1771561/60000000              | 9 to 2  | 11/2             | 19487171/120000000         |
| Total exactly 8   | 2.7066%     | 19487171/720000000            | 4 to 1  | 5                | 19487171/144000000         |
| Total exactly 9   | 2.3818%     | 214358881/9000000000          | 7 to 2  | 9/2              | 214358881/2000000000       |
| Total exactly 10  | 1.9650%     | 2357947691/120000000000       | 3 to 1  | 4                | 2357947691/30000000000     |
| Total exactly 11  | 0.8854%     | 5312474867/600000000000       | 5 to 1  | 6                | 5312474867/100000000000    |
| Total exactly 12  | 0.4592%     | 165311670611/3600000000000    | 5 to 1  | 6                | 165311670611/6000000000000 |
| Any other outcome | 82.6990%    | 29771629856069/36000000000000 | loses   | 0                | 0                          |

- **Total return (RTP):** 17307296056493/18000000000000 (96.1516%), the sum of the contributions.
- **House edge:** 692703943507/18000000000000 (3.8484%).
- **Variance** of the net result per unit staked: 1494642153505545850952540951/3240000000000000000000000 (4.6131); **standard deviation** 2.1478.
- **Hit frequency** (the wager wins): 6228370143931/36000000000000 (17.3010%).

Reproduced by packages/engine/src/games/moving-target/moving-target.test.ts › Moving Target — exact math › the real game over every roll and card sequence › breaks every bet down by target, adding up to its declared figures, and finds the most a round can pay.

4.2 First Card (side)

**Wins** when  $c_1 = T$  (possible only for  $T \leq 10$ ). **Method:** a memoised recursion over the running total and the card count, written from the rules alone, checked against the closed forms of bets.ts and against exact enumeration of the production game over  $\Omega$ ; all three agree as fractions.

By target, weighted by the dice:

| Target      | P(target) | P(win   target) | Pays   | Return given target | House edge given target | P(target and win) | Contribution to RTP |
|-------------|-----------|-----------------|--------|---------------------|-------------------------|-------------------|---------------------|
| 2           | 1/36      | 1/10 (10.00%)   | 9 to 1 | 100.000%            | 0.000%                  | 0.2778%           | 2.7778%             |
| 3           | 1/18      | 1/10 (10.00%)   | 9 to 1 | 100.000%            | 0.000%                  | 0.5556%           | 5.5556%             |
| 4           | 1/12      | 1/10 (10.00%)   | 9 to 1 | 100.000%            | 0.000%                  | 0.8333%           | 8.3333%             |
| 5           | 1/9       | 1/10 (10.00%)   | 9 to 1 | 100.000%            | 0.000%                  | 1.1111%           | 11.1111%            |
| 6           | 5/36      | 1/10 (10.00%)   | 9 to 1 | 100.000%            | 0.000%                  | 1.3889%           | 13.8889%            |
| 7           | 1/6       | 1/10 (10.00%)   | 9 to 1 | 100.000%            | 0.000%                  | 1.6667%           | 16.6667%            |
| 8           | 5/36      | 1/10 (10.00%)   | 9 to 1 | 100.000%            | 0.000%                  | 1.3889%           | 13.8889%            |
| 9           | 1/9       | 1/10 (10.00%)   | 9 to 1 | 100.000%            | 0.000%                  | 1.1111%           | 11.1111%            |
| 10          | 1/12      | 1/10 (10.00%)   | 9 to 1 | 100.000%            | 0.000%                  | 0.8333%           | 8.3333%             |
| 11          | 1/18      | 0 (0.00%)       | 9 to 1 | 0.000%              | 100.000%                | 0.0000%           | 0.0000%             |
| 12          | 1/36      | 0 (0.00%)       | 9 to 1 | 0.000%              | 100.000%                | 0.0000%           | 0.0000%             |
| All targets |           |                 |        |                     |                         |                   | 91.6667%            |

As an outcome table:

| Outcome                      | Probability | Exactly | Pays   | Returns per unit | Contribution to RTP |
|------------------------------|-------------|---------|--------|------------------|---------------------|
| The first card is the target | 9.1667%     | 11/120  | 9 to 1 | 10               | 11/12               |
| Any other outcome            | 90.8333%    | 109/120 | loses  | 0                | 0                   |

- **Total return (RTP):** 11/12 (91.6667%), the sum of the contributions.
- **House edge:** 1/12 (8.3333%).
- **Variance** of the net result per unit staked: 1199/144 (8.3264); **standard deviation** 2.8855.
- **Hit frequency** (the wager wins): 11/120 (9.1667%).

Reproduced by packages/engine/src/games/moving-target/moving-target.test.ts › Moving Target — exact math › the real game over every roll and card sequence › breaks every bet down by target, adding up to its declared figures, and finds the most a round can pay.

4.3 3+ Cards (side)

**Wins** when  $K \geq 3$ , that is  $c_1 + c_2 < T$ . **Method:** a memoised recursion over the running total and the card count, written from the rules alone, checked against the closed forms of bets.ts and against exact enumeration of the production game over  $\Omega$ ; all three agree as fractions.

By target, weighted by the dice:

| Target      | P(target) | P(win   target) | Pays   | Return given target | House edge given target | P(target and win) | Contribution to RTP |
|-------------|-----------|-----------------|--------|---------------------|-------------------------|-------------------|---------------------|
| 2           | 1/36      | 0 (0.00%)       | 4 to 1 | 0.000%              | 100.000%                | 0.0000%           | 0.0000%             |
| 3           | 1/18      | 1/100 (1.00%)   | 4 to 1 | 5.000%              | 95.000%                 | 0.0556%           | 0.2778%             |
| 4           | 1/12      | 3/100 (3.00%)   | 4 to 1 | 15.000%             | 85.000%                 | 0.2500%           | 1.2500%             |
| 5           | 1/9       | 3/50 (6.00%)    | 4 to 1 | 30.000%             | 70.000%                 | 0.6667%           | 3.3333%             |
| 6           | 5/36      | 1/10 (10.00%)   | 4 to 1 | 50.000%             | 50.000%                 | 1.3889%           | 6.9444%             |
| 7           | 1/6       | 3/20 (15.00%)   | 4 to 1 | 75.000%             | 25.000%                 | 2.5000%           | 12.5000%            |
| 8           | 5/36      | 21/100 (21.00%) | 4 to 1 | 105.000%            | −5.000%                 | 2.9167%           | 14.5833%            |
| 9           | 1/9       | 7/25 (28.00%)   | 4 to 1 | 140.000%            | −40.000%                | 3.1111%           | 15.5556%            |
| 10          | 1/12      | 9/25 (36.00%)   | 4 to 1 | 180.000%            | −80.000%                | 3.0000%           | 15.0000%            |
| 11          | 1/18      | 9/20 (45.00%)   | 4 to 1 | 225.000%            | −125.000%               | 2.5000%           | 12.5000%            |
| 12          | 1/36      | 11/20 (55.00%)  | 4 to 1 | 275.000%            | −175.000%               | 1.5278%           | 7.6389%             |
| All targets |           |                 |        |                     |                         |                   | 89.5833%            |

As an outcome table:

| Outcome             | Probability | Exactly | Pays   | Returns per unit | Contribution to RTP |
|---------------------|-------------|---------|--------|------------------|---------------------|
| Three cards or more | 17.9167%    | 43/240  | 4 to 1 | 5                | 43/48               |
| Any other outcome   | 82.0833%    | 197/240 | loses  | 0                | 0                   |

- **Total return (RTP):** 43/48 (89.5833%), the sum of the contributions.
- **House edge:** 5/48 (10.4167%).
- **Variance** of the net result per unit staked: 8471/2304 (3.6766); **standard deviation** 1.9175.
- **Hit frequency** (the wager wins): 43/240 (17.9167%).

Reproduced by `packages/engine/src/games/moving-target/moving-target.test.ts` › *Moving Target — exact math* › the real game over every roll and card sequence › breaks every bet down by target, adding up to its declared figures, and finds the most a round can pay.

## 5. Progressive analysis

Not applicable: Moving Target has no progressive wager.

## 6. Finite-shoe effects

Two tests measure the 6-deck shoe against the declared, infinite-shoe figures. The first round after a shuffle is computed exactly, by a recursion over the cards drawn without replacement from a full shoe (24 cards of each value): a second card of a value already drawn is less likely, so totals reached through pairs get rarer. The long run is simulated on the production game, with its cut card after 180 cards and reshuffles: over whole shoes the rounds interact, since each round takes cards whose mix depends on the target it chased, and the number of rounds a shoe deals depends on its cards.

| Wager      | House edge, declared | First round, exact | Long run, observed | Long-run shift | Standard error |
|------------|----------------------|--------------------|--------------------|----------------|----------------|
| Exact Hit  | 3.848%               | 3.978%             | 3.885%             | +0.036 pp      | 0.016 pp       |
| First Card | 8.333%               | 8.333%             | 8.391%             | +0.058 pp      | 0.021 pp       |
| 3+ Cards   | 10.417%              | 10.617%            | 10.302%            | −0.114 pp      | 0.014 pp       |

Exact Hit by target (house edge; a positive shift is dearer for the player):

| Target | Declared | First round, exact | Long run, observed | Long-run shift | Standard error |
|--------|----------|--------------------|--------------------|----------------|----------------|
| 2      | 6.50%    | 6.82%              | 6.59%              | +0.09 pp       | 0.118 pp       |
| 3      | 3.20%    | 3.22%              | 2.93%              | −0.27 pp       | 0.082 pp       |
| 4      | 6.83%    | 7.11%              | 7.06%              | +0.23 pp       | 0.061 pp       |
| 5      | 4.83%    | 4.87%              | 4.70%              | −0.13 pp       | 0.051 pp       |
| 6      | 3.37%    | 3.62%              | 3.49%              | +0.12 pp       | 0.044 pp       |
| 7      | 2.56%    | 2.60%              | 2.49%              | −0.08 pp       | 0.038 pp       |
| 8      | 2.56%    | 2.78%              | 2.79%              | +0.22 pp       | 0.039 pp       |
| 9      | 3.54%    | 3.56%              | 3.50%              | −0.04 pp       | 0.041 pp       |
| 10     | 5.68%    | 5.84%              | 5.86%              | +0.18 pp       | 0.043 pp       |
| 11     | 4.38%    | 4.38%              | 4.27%              | −0.10 pp       | 0.069 pp       |
| 12     | 0.81%    | 1.06%              | 0.96%              | +0.15 pp       | 0.099 pp       |

Which figures a table returns depends on how it deals: an RNG table that draws every card independently returns the declared figures; one that reshuffles a virtual 6-deck shoe before every round, as a continuous shuffler does, returns the first-round column; a shoe dealt down to the cut card, as at a live table, returns the long-run column. No wager's edge moves by more than 0.25 pp (the test's bound). *Reproduced by* `packages/engine/src/games/moving-target/finite-shoe.test.ts` › *Moving Target — the first round of a fresh six-deck shoe, exactly* › *measures every bet, and Exact Hit by target, on the first round of a fresh shoe;* `packages/engine/src/games/moving-target/six-deck.math.test.ts` › *Moving Target — the six-deck shoe against the declared (infinite-shoe) figures* › *shifts no edge by more than a quarter of a point, and reports each shift.*

## 7. Simulation verification

On the infinite shoe, against the declared figures:

|           |                                                                                                                                                              |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rounds    | 40,055,852                                                                                                                                                   |
| Generator | xoshiro128** 1.1 (Blackman & Vigna), its state expanded from the seed by SplitMix64; a string seed is first hashed with FNV-1a (64-bit) over its UTF-8 bytes |
| Seed      | moving-target/infinite-shoe                                                                                                                                  |

|               |                                                                                                                                                                                                                 |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Rounds</b> | <b>40,055,852</b>                                                                                                                                                                                               |
| Cards         | an infinite shoe (every card drawn independently and uniformly from one deck of ranks 1 to 10 in 4 suits)                                                                                                       |
| Stakes        | every wager at its minimum each round: Exact Hit 0.50, First Card 0.50 and 3+ Cards 0.50                                                                                                                        |
| Criterion     | each RTP within $\pm 0.15$ pp of the declared figure and each hit frequency within 3.29 binomial standard errors; the round count makes $\pm 0.15$ pp equal to 3.29 standard errors for the most volatile wager |

| Figure                    | Observed | Expected | Difference | Standard error | 95% confidence interval | Allowed        | Result |
|---------------------------|----------|----------|------------|----------------|-------------------------|----------------|--------|
| Exact Hit, RTP            | 96.153%  | 96.152%  | +0.001 pp  | 0.034 pp       | 96.086% to 96.219%      | $\pm 0.150$ pp | pass   |
| Exact Hit, hit frequency  | 17.300%  | 17.301%  | -0.001 pp  | 0.006 pp       | 17.289% to 17.312%      | $\pm 0.020$ pp | pass   |
| First Card, RTP           | 91.668%  | 91.667%  | +0.001 pp  | 0.046 pp       | 91.578% to 91.757%      | $\pm 0.150$ pp | pass   |
| First Card, hit frequency | 9.167%   | 9.167%   | 0.000 pp   | 0.005 pp       | 9.158% to 9.176%        | $\pm 0.015$ pp | pass   |
| 3+ Cards, RTP             | 89.578%  | 89.583%  | -0.006 pp  | 0.030 pp       | 89.518% to 89.637%      | $\pm 0.150$ pp | pass   |
| 3+ Cards, hit frequency   | 17.916%  | 17.917%  | -0.001 pp  | 0.006 pp       | 17.904% to 17.927%      | $\pm 0.020$ pp | pass   |

Reproduced by `packages/engine/src/games/moving-target/moving-target.math.test.ts` > *Moving Target — Monte Carlo against the declared figures (infinite shoe)* > *lands every bet within  $\pm 0.15$  pp of its declared RTP.*

On the 6-deck shoe, measuring the shift of section six rather than checking a declared figure:

|               |                                                                                                                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Rounds</b> | <b>183,929,931</b>                                                                                                                                                                                                             |
| Generator     | xoshiro128** 1.1 (Blackman & Vigna), its state expanded from the seed by SplitMix64; a string seed is first hashed with FNV-1a (64-bit) over its UTF-8 bytes                                                                   |
| Seed          | moving-target/six-deck-shoe                                                                                                                                                                                                    |
| Cards         | the 6-deck shoe of 240 cards, ranks 1 to 10, with its cut card after 180 cards (penetration 75%) and reshuffles, as the production game deals it                                                                               |
| Stakes        | every wager at its minimum each round: Exact Hit 0.50, First Card 0.50 and 3+ Cards 0.50                                                                                                                                       |
| Criterion     | no wager's house edge shifted by more than 0.25 pp from the declared figure, and no target's by more than that plus 3.29 standard errors; the round count measures each wager's shift to $\pm 0.07$ pp at 3.29 standard errors |

| Figure                 | Observed | Expected | Difference | Standard error | 95% confidence interval | Allowed        | Result |
|------------------------|----------|----------|------------|----------------|-------------------------|----------------|--------|
| Exact Hit, house edge  | 3.885%   | 3.848%   | +0.036 pp  | 0.016 pp       | 3.854% to 3.916%        | $\pm 0.250$ pp | pass   |
| First Card, house edge | 8.391%   | 8.333%   | +0.058 pp  | 0.021 pp       | 8.349% to 8.433%        | $\pm 0.250$ pp | pass   |
| 3+ Cards, house edge   | 10.302%  | 10.417%  | -0.114 pp  | 0.014 pp       | 10.275% to 10.330%      | $\pm 0.250$ pp | pass   |

The 95% confidence interval is the observed value  $\pm 1.96$  standard errors, and at that level about one figure in twenty falls outside its interval by chance. The tests' own criterion, "Allowed" above, is wider: 3.29 standard errors, or a fixed tolerance. *Reproduced by* `packages/engine/src/games/moving-target/six-deck.math.test.ts` > *Moving Target — the six-deck shoe against the declared (infinite-shoe) figures* > *shifts no edge by more than a quarter of a point, and reports each shift.*

## 8. Assumptions and limitations

**Independence.** The dice and the cards are independent: the RNG draws each die and each card separately (a live table uses separate devices), and nothing about the dice changes the shoe. Within a round the cards are dealt from one shoe, which the finite-shoe figures account for.

**Card counting.** The cards are dealt face up, so a player who tracks them knows the shoe's composition before every round. A seeded run of 5,000,000 rounds of the production game (seed `moving-target/counting`, 50,291 shoes) computes each wager's exact expectation for the composition before every round: the chance that some first  $k$  cards of the shoe add up to the target is a sum over the multisets of values that do, each with its hypergeometric chance, which the test checks against a direct recursion.

| Wager      | Rounds favouring a perfect counter | Counter's edge in them | Break-even bet spread |
|------------|------------------------------------|------------------------|-----------------------|
| Exact Hit  | 11.07%                             | 2.8%                   | 1 to 13.6             |
| First Card | 0.85%                              | 1.7%                   | 1 to 580.0            |
| 3+ Cards   | 11.55%                             | 7.4%                   | 1 to 13.0             |

The break-even spread is how much more a perfect counter must stake in the favourable rounds than in the others (one unit) to break even. Exact Hit and 3+ Cards can be counted at the table's penetration, First Card barely. A live table should use a continuous shuffling machine or shuffle much earlier; an RNG table that reshuffles every round is immune. *Reproduced by* `packages/engine/src/games/moving-target/counting.math.test.ts` > *Moving Target — card counting on the six-deck shoe* > *measures how often the cards left favour each bet, and the spread a counter needs.*

**Rounding.** Winnings are paid to the cent, rounded down (the engine's `winnings()`); the declared figures assume exact pays. Every odds in these games has a denominator of one or two, so a wager in multiples of the minimum pays exactly and the rounding never applies at the table's stakes.

**No player or dealer error** is modelled, nor any irregularity: the figures are those of the rules as the engine settles them.

**Randomness.** The analysis assumes a certified RNG: every die face and every card position equally likely and independent. The engine draws integers by rejection sampling, without modulo bias, and shuffles by Fisher-Yates in a specified order.

## 9. Appendix

The full enumeration of 71,469 outcomes does not fit a page; the tests hold it. `moving-target.test.ts` runs the production game over every roll and card sequence and checks each wager's RTP, hit frequency, lines and variance against the recursion, and its record holds the per-target figures above as exact fractions. The number of card sequences by target:

| Target | Rolls | Card sequences |
|--------|-------|----------------|
| 2      | 1     | 19             |
| 3      | 2     | 37             |
| 4      | 3     | 73             |

| Target | Rolls | Card sequences |
|--------|-------|----------------|
| 5      | 4     | 145            |
| 6      | 5     | 289            |
| 7      | 6     | 577            |
| 8      | 5     | 1,153          |
| 9      | 4     | 2,305          |
| 10     | 3     | 4,609          |
| 11     | 2     | 9,217          |
| 12     | 1     | 18,424         |

**Records.** The tests recorded these figures ( `packages/engine/src/games/moving-target/results/` ):

| Record                          | Test                                                                                                                                                                                                                                                               |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>counting.json</code>      | <code>packages/engine/src/games/moving-target/counting.math.test.ts</code> › Moving Target — card counting on the six-deck shoe › measures how often the cards left favour each bet, and the spread a counter needs                                                |
| <code>exact.json</code>         | <code>packages/engine/src/games/moving-target/moving-target.test.ts</code> › Moving Target — exact math › the real game over every roll and card sequence › breaks every bet down by target, adding up to its declared figures, and finds the most a round can pay |
| <code>first-round.json</code>   | <code>packages/engine/src/games/moving-target/finite-shoe.test.ts</code> › Moving Target — the first round of a fresh six-deck shoe, exactly › measures every bet, and Exact Hit by target, on the first round of a fresh shoe                                     |
| <code>infinite-shoe.json</code> | <code>packages/engine/src/games/moving-target/moving-target.math.test.ts</code> › Moving Target — Monte Carlo against the declared figures (infinite shoe) › lands every bet within $\pm 0.15$ pp of its declared RTP                                              |
| <code>six-deck-shoe.json</code> | <code>packages/engine/src/games/moving-target/six-deck.math.test.ts</code> › Moving Target — the six-deck shoe against the declared (infinite-shoe) figures › shifts no edge by more than a quarter of a point, and reports each shift                             |

# Mirror — Rules of Play

*Your dice against the dealer's cards, hand for hand.*

## 1. Game and document

|            |                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------|
| Game       | Mirror, an original table game of the Roll & Deal family                                                            |
| Version    | 0.1.0                                                                                                               |
| Document   | Rules of Play, revision 3                                                                                           |
| Date       | 2026-09-30                                                                                                          |
| Author     | Jonathas Costa                                                                                                      |
| Companion  | Math Report ( docs/math/mirror.md )                                                                                 |
| Demo table | <a href="https://jonathascosta.github.io/CasinoGames/mirror">https://jonathascosta.github.io/CasinoGames/mirror</a> |

### Revision history

| Revision | Date       | Author         | Change                                                                                                                                                                                         |
|----------|------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | 2026-09-30 | Jonathas Costa | First game sheet: rules, declared math, exact and Monte Carlo tests. The progressive meter's economics and card counting measured.                                                             |
| 2        | 2026-09-30 | Jonathas Costa | Renamed from its working title, with English names for the game and its bets. The progressive bet is Double Sixes.                                                                             |
| 3        | 2026-09-30 | Jonathas Costa | Reformatted for submission as two documents, Rules of Play and Math Report, generated from the code with every figure traced to the engine or to a test; card counting measured on every game. |

Money is written in the table's currency with two decimals, as the demo shows it (the demo plays with virtual chips).

## 2. Objective

The player's two dice and the dealer's two cards are two hands, read the same way: a pair above any non-pair, then by sum and by the higher value. The main wager, Mirror, wins when the dice outrank the cards, and side wagers pay on the two hands mirroring each other, one of them with a progressive meter.

## 3. Equipment

- **Dice.** Two standard six-sided dice, faces 1 to 6, thrown by the player (by the round's shooter at a live table: see section nine). The two dice are the player's **hand**.
- **Shoe.** 6 decks, each of the A to the 6 in 4 suits (24 cards): 144 cards in all, made from standard decks with every card above the 6 removed. The ace counts 1 and every other card its number. The dealer's two cards are the dealer's hand.
- **Cut card.** It goes in with 36 cards behind it, a penetration of 75%: 108 cards are dealt before it comes out. When it comes out, the round in progress is completed, and the whole shoe is shuffled before the next round.
- **Progressive meter.** A display, visible to every seat, of the Double Sixes meter (section eight).
- **Cosmetic.** Suits never count, only values. Which die is which, and the order of the two cards, do not matter: a hand is read by its two values.
- **Also.** The layout (section four), chips, a discard rack and, at a live table, a dice cup or shaker.

## 4. Table layout

Each player position has one spot for the main wager, Mirror, in the centre, one for each fixed-odds side wager (Tie, Equal Sums, Pair vs Pair and Perfect Mirror) and one for the progressive wager, Double Sixes. The progressive meter hangs across the top of the layout. The felt prints:

| Spot           | Wins when                                                               | Pays                                | A 1.00 wager wins                        |
|----------------|-------------------------------------------------------------------------|-------------------------------------|------------------------------------------|
| Mirror         | the dice outrank the cards (a tie loses)                                | 1 to 1                              | 1.00                                     |
| Tie            | the two hands rank exactly equal: the same pair, or the same two values | 17 to 1                             | 17.00                                    |
| Equal Sums     | the dice and the cards add up to the same total                         | 7 to 1                              | 7.00                                     |
| Pair vs Pair   | both hands are pairs                                                    | 30 to 1                             | 30.00                                    |
| Perfect Mirror | both hands are the same pair                                            | 200 to 1                            | 200.00                                   |
| Double Sixes   | both hands are 6-6                                                      | 1000 to 1 plus a share of the meter | 1,000.00 plus stake + 25.00 of the meter |

The demo shows this layout at <https://jonathascosta.github.io/CasinoGames/mirror>. At the top is the mirror: the dealer's cards above the glass, the dice below it, each hand labelled as the table reads it (PAIR 4s, SUM 9 HIGH 6), and the glass tilting toward the winner. Below is the felt, with the meter across its top, Mirror in the centre, Double Sixes under it and two side wagers on each side, then the chips, the balance, the total wagered and the Clear, Roll and Autoplay buttons, with Rules and Paytable buttons beside the balance; an RTP monitor below the table compares the rounds played with the declared figures. The lobby shows the live meter on the table's card.

## 5. Wagers

Every wager is placed before the roll and settled once both hands are known. The main wager comes first, the fixed-odds side wagers next and the progressive wager last.

### 5.1 Mirror (main wager)

- **When.** Before the roll. It can be placed alone, and every side wager needs it.
- **Limits.** 0.50 to 250.00.
- **Wins** when the dice outrank the cards (section seven).

- **Pays** 1 to 1.
- **Pushes** never: a tie loses.
- **Loses** when the cards outrank the dice or the hands tie.

## 5.2 Tie (side wager)

- **When.** Before the roll, only with a Mirror wager.
- **Limits.** 0.50 to 25.00.
- **Wins** when the two hands rank exactly equal: the same pair, or the same two values.
- **Pays** 17 to 1.
- **Pushes** never.
- **Loses** otherwise.

## 5.3 Equal Sums (side wager)

- **When.** Before the roll, only with a Mirror wager.
- **Limits.** 0.50 to 25.00.
- **Wins** when the dice and the cards add up to the same total.
- **Pays** 7 to 1.
- **Pushes** never.
- **Loses** otherwise.

## 5.4 Pair vs Pair (side wager)

- **When.** Before the roll, only with a Mirror wager.
- **Limits.** 0.50 to 25.00.
- **Wins** when both hands are pairs.
- **Pays** 30 to 1.
- **Pushes** never.
- **Loses** otherwise.

## 5.5 Perfect Mirror (side wager)

- **When.** Before the roll, only with a Mirror wager.
- **Limits.** 0.50 to 25.00.
- **Wins** when both hands are the same pair.
- **Pays** 200 to 1.
- **Pushes** never.
- **Loses** otherwise.

## 5.6 Double Sixes (progressive side wager)

- **When.** Before the roll, only with a Mirror wager. 10% of the stake goes into the meter as the wager is accepted, win or lose.
- **Limits.** 0.50 to 25.00.
- **Wins** when both hands are 6-6: the dice a pair of 6s and the dealer's two cards both 6s.
- **Pays** 1000 to 1 on the stake, plus the stake's share of the meter: stake ÷ 25.00 of it, so the whole meter at the 25.00 maximum and 2% of it at the 0.50 minimum.
- **Pushes** never.
- **Loses** otherwise; its contribution to the meter stays in the meter.

## 6. Sequence of play

---

1. **Wagers.** The dealer invites wagers. Each player places Mirror and any side wagers, within the limits, and the dealer closes betting ("No more bets"). The meter is raised by 10% of every Double Sixes stake.
2. **Shuffle, if due.** If the cut card came out during the previous round, the dealer shuffles the whole shoe, has it cut and places the cut card with 36 cards behind it before the round begins.
3. **Roll.** The player rolls both dice. The dealer announces the player's hand ("PAIR 4s", "SUM 9 HIGH 6").
4. **Deal.** The dealer deals two cards from the shoe, face down, then turns them over one at a time and announces the dealer's hand.
5. **Compare.** The dealer compares the two hands (section seven) and announces the winner, or a tie, and the side wagers that win.
6. **Settle.** The dealer settles every wager (section eight), pays any Double Sixes win from the meter, clears the cards and returns the dice.

## 7. Hand ranking

---

A hand is two values from 1 to 6, and the dice and the cards are ranked the same way:

1. A **pair** beats any non-pair, and a higher pair beats a lower one: from PAIR 6s down to PAIR 1s.
2. Between two non-pairs, the higher **sum** wins; on equal sums, the higher of the two values wins. The table calls such a hand by its sum and its higher value: SUM 11 HIGH 6 is a 6 and a 5.
3. Two hands still level hold the same two values: a **tie**, which the house wins on the main wager.

From the strongest down: PAIR 6s, PAIR 5s, PAIR 4s, PAIR 3s, PAIR 2s, PAIR 1s, SUM 11 HIGH 6, SUM 10 HIGH 6, SUM 9 HIGH 6, SUM 9 HIGH 5, SUM 8 HIGH 6, SUM 8 HIGH 5, SUM 7 HIGH 6, SUM 7 HIGH 5, SUM 7 HIGH 4, SUM 6 HIGH 5, SUM 6 HIGH 4, SUM 5 HIGH 4, SUM 5 HIGH 3, SUM 4 HIGH 3, SUM 3 HIGH 2. Even PAIR 1s beats SUM 11 HIGH 6.

Worked examples:

- Dice 6-3 (SUM 9 HIGH 6) against cards 5-4 (SUM 9 HIGH 5): equal sums, and the dice's higher value is higher. The dice win: Mirror wins 1 to 1; Equal Sums wins 7 to 1; Tie, Pair vs Pair, Perfect Mirror and Double Sixes lose.
- Dice 1-1 (PAIR 1s) against cards 6-5 (SUM 11 HIGH 6): a pair beats any non-pair. The dice win: Mirror wins 1 to 1; Tie, Equal Sums, Pair vs Pair, Perfect Mirror and Double Sixes lose.
- Dice 5-2 (SUM 7 HIGH 5) against cards 5-2 (SUM 7 HIGH 5): the same two values, a tie. Tie wins 17 to 1; Equal Sums wins 7 to 1; Mirror, Pair vs Pair, Perfect Mirror and Double Sixes lose.
- Dice 4-4 (PAIR 4s) against cards 4-4 (PAIR 4s): the same pair, a tie. Tie wins 17 to 1; Equal Sums wins 7 to 1; Pair vs Pair wins 30 to 1; Perfect Mirror wins 200 to 1; Mirror and Double Sixes lose.

## 8. Settlement procedure

---

- The dealer settles once both hands are known: first collecting every losing wager, then paying the winners in the felt's order, Mirror first, the fixed-odds side wagers next and Double Sixes last. A winning wager is paid at its odds and keeps its stake.
- **The meter.** It starts at its seed, 5,000.00, and grows by 10% of every Double Sixes stake, credited as the wager is accepted: the house funds it out of the stakes it holds, and a winning wager is still paid on its whole stake.
- **A hit** pays 1000 to 1 on the stake from the table's bank, plus the stake ÷ 25.00 share of the meter, rounded down to the cent, from the meter. The round's own contribution is already in the meter when the share is taken.
- **After a hit** the meter is decreased by the share paid and keeps the rest, down to the fraction of a cent. If it falls below the seed, the house tops it back up to 5,000.00.
- **Example.** 5.00 on Double Sixes with the meter at 6,000.00: the stake takes it to 6,000.50. Both hands are 6-6: the wager wins 5,000.00 at 1000 to 1 plus 1,200.10 from the meter, and keeps its stake. The meter keeps 4,800.40, which the house tops up to 5,000.00.
- A fixed pay that is not a whole number of cents is rounded down to the cent; with these odds it never is.

## 9. Physical and live-dealer adaptation

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- **Dice.** The player throws from a dice cup or by hand, across the layout, so that both dice tumble and come to rest on the felt. At a table with several players, one shooter throws per round, in turn, and that one roll and one pair of cards settle every seat; an automatic shaker, as used for Sic Bo, can replace the throw.
- **Calls.** The dealer announces both hands by the table's names ("SUM 9 HIGH 6 against SUM 9 HIGH 5: the dice win").
- **Shoe.** Prepare 6 decks of A to 6 (144 cards), shuffle, offer the cut and place the cut card with 36 cards behind it. With two cards a round, a shoe deals 54 rounds.
- **Meter.** A meter display driven by the table's controller (or a network of tables sharing one meter), credited with every Double Sixes contribution as betting closes. A hit is verified by a supervisor before the meter is paid, as for any progressive.
- **Several winners.** When several seats win Double Sixes in the same round, each seat is due stake ÷ 25.00 of the meter as it stood before the round's payments; if those shares add up to more than the whole meter, the meter is divided among the seats in proportion to their stakes.
- **Card counting.** The cards are dealt openly, and Mirror and Double Sixes can be counted at this penetration: the Math Report measures by how much. A live table should use a continuous shuffling machine, or shuffle much earlier.
- **The RNG version** draws the dice and the cards from a certified random number generator, deals from a virtual shoe with the same cut card (or reshuffles it every round), keeps the meter to a millionth of a cent and settles at once; the player's tap or throw only animates the dice. It has no physical irregularities. Every rule, limit and pay is the same.

## 10. Irregularities

---

- **Die off the table.** A die that leaves the layout (off the table, into the chip rail or the dealer's bank) is no roll: the dealer calls it, checks the dice and the shooter rolls again. Wagers stay as they are.
- **Cocked die.** A die that comes to rest tilted, against a chip, the rail, a card or the other die, so that no face is plainly uppermost, is no roll: the dice are rolled again. A die lying flat on a chip is read as it lies.
- **Dice thrown too early.** Dice thrown before the dealer closes betting are no roll.
- **Misread result.** The dice and cards as they lie decide the round: a misread call is corrected before the round is settled, and a settlement error is corrected before the next round begins.
- **Exposed card.** A card exposed before the roll is burned to the discards, and the round proceeds with the next cards. One of the round's two cards dealt face up instead of face down stands: facing them down is only presentation.
- **Mis-deal.** Only the first two cards dealt after the roll count; any further card, or a card dealt before the roll, is burned.
- **Meter fault.** If the meter display fails, the controller's record of the meter decides the share paid.
- **Cards short of the cut card.** It cannot happen as the shoe is prepared: a round uses at most two cards, and 36 cards sit behind the cut card. If the shoe were nonetheless to run out during a round (a cut card placed too deep, cards missing), the dealer shuffles the discards of earlier rounds, never the cards of the round in progress, and completes the round from them; the whole shoe is then shuffled before the next round. This is what the RNG version does.

## To confirm

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These rulings are standard casino practice, not rules the game's code decides. Confirm each one, or replace it with the house's own, before a live table opens.

- Settlement order: losing wagers collected first, then the winners paid, main wager first.
- Dice off the table and cocked dice are no roll, rather than read where they lie.
- At a table with several players, one shooter per round, in turn (or an automatic shaker), whose roll settles every seat.
- No card is burned after the shuffle, as in the RNG version; a burn card would not change the rules.
- Several Double Sixes winners in one round share the meter in proportion to their stakes when their shares exceed it.
- A supervisor verifies every Double Sixes hit before the meter is paid.

- An exposed card is burned; the round's card dealt face up in error stands; extra cards are burned.
- Counting: a continuous shuffler, or a much earlier shuffle, at a live table.

# Mirror — Math Report

Your dice against the dealer's cards, hand for hand.

## 1. Cover

| Game      | Mirror, an original table game of the Roll & Deal family                                                                         |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|
| Version   | 0.1.0                                                                                                                            |
| Document  | Math Report, revision 3                                                                                                          |
| Date      | 2026-09-30                                                                                                                       |
| Author    | Jonathas Costa                                                                                                                   |
| Code      | commit 47bab22a941c983cd3e18fd6989ebb7a1e6479b5 (2026-09-30): the last change to the engine, its tests or their recorded results |
| Figures   | docs/results.json, written with this report by tools/generate-docs.ts                                                            |
| Companion | Rules of Play ( docs/rules/mirror.md )                                                                                           |

### Revision history

| Revision | Date       | Author         | Change                                                                                                                                                                                         |
|----------|------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | 2026-09-30 | Jonathas Costa | First game sheet: rules, declared math, exact and Monte Carlo tests. The progressive meter's economics and card counting measured.                                                             |
| 2        | 2026-09-30 | Jonathas Costa | Renamed from its working title, with English names for the game and its bets. The progressive bet is Double Sixes.                                                                             |
| 3        | 2026-09-30 | Jonathas Costa | Reformatted for submission as two documents, Rules of Play and Math Report, generated from the code with every figure traced to the engine or to a test; card counting measured on every game. |

**How to read this report.** Every figure is read from `docs/results.json`, which the generator builds from the game's `mathSummary()` (the declared figures) and from the figures its tests record (`packages/engine/src/games/mirror/results/`); the generator fails if a document holds a number that is not in it. Each table names the test that reproduces it, as file > suite > test: `pnpm test` runs the exact tests and `pnpm test:math` the simulations, and each fails if it no longer reproduces its recorded figures. Returns are per unit staked, the stake included; exact figures are fractions, rounded percentages beside them.

## 2. Summary

| Wager                 | RTP      | House edge | Hit frequency | SD per unit | Max payout | Max exposure at max bet |
|-----------------------|----------|------------|---------------|-------------|------------|-------------------------|
| Mirror (main)         | 94.9074% | 5.0926%    | 47.4537%      | 0.9987      | 1×         | 250.00 (on 250.00)      |
| Tie (side)            | 91.6667% | 8.3333%    | 5.0926%       | 3.9572      | 17×        | 425.00 (on 25.00)       |
| Equal Sums (side)     | 90.1235% | 9.8765%    | 11.2654%      | 2.5294      | 7×         | 175.00 (on 25.00)       |
| Pair vs Pair (side)   | 86.1111% | 13.8889%   | 2.7778%       | 5.0944      | 30×        | 750.00 (on 25.00)       |
| Perfect Mirror (side) | 93.0556% | 6.9444%    | 0.4630%       | 13.6446     | 200×       | 5,000.00 (on 25.00)     |
| Double Sixes (side)   | 87.2377% | 12.7623%   | 0.0772%       | 33.3482     | 1,200×     | 30,000.00 (on 25.00)    |

- **RTP** and **house edge** are per unit staked, pushes included. **Hit frequency** is the chance that the wager wins in a round. **SD per unit** is the standard deviation of the net result per unit staked.
- **Max payout** is the largest net win per unit staked, the stake being returned on top. **Max exposure at max bet** is the most a single wager at its maximum can win in a round, found by running the real game over every outcome at the table maximums; all the wagers of one position together can win at most 36,100.00 in a round.
- **Double Sixes** pays a share of the progressive meter on top of its fixed odds. Its RTP excludes the meter's seed, which the house funds: the fixed pays plus the contributions. Its SD per unit, max payout and max exposure are taken with the meter at its seed; above it they grow with the meter.

Reproduced by `packages/engine/src/games/mirror/mirror.test.ts` > *Mirror — the game over every draw of the dice and an infinite shoe* > adds up each bet's outcomes to its declared figures, and finds the most a round can pay. The declared figures are `mathSummary()` (`bets.ts`); the test holds them equal to the enumeration.

## 3. Game model

- **Sample space.**  $\Omega = D \times C^2$ .  $D$  is the ordered pair of dice ( $d_1, d_2$ ), uniform on 1 to 6;  $C^2$  the ordered pair of card values ( $c_1, c_2$ ). Each hand is read by its two values: a pair ranks above every non-pair, by its value; non-pairs rank by sum, then by the higher value, so only identical hands tie. The exact test enumerates 36 rolls  $\times$  576 ordered pairs of cards from one deck's 24 cards: 20,736 outcomes.
- **Card supply.** The declared figures take the two cards independent and uniform over the values: an infinite shoe, as an RNG game that draws each card independently deals them. On the table's 6-deck shoe a pair of cards is rarer (once a card is out, fewer of its value are left), and section six gives those figures exactly: with two cards a round and 54 rounds a shoe whatever the cards, the two cards of any round are a uniform draw of two cards from the full shoe.
- **Decisions.** None: every wager is settled by the two hands.

## 4. Per-wager analysis

### 4.1 Mirror (main)

**Wins** when `rank(dice) > rank(cards)`. **Method:** exact enumeration of the production game over  $\Omega$ . The ranking is enumerated in the appendix: the dice outrank the cards with chance 205/432 (47.4537%), tie with chance 11/216 (5.0926%) and lose with chance 205/432 (47.4537%).

| Outcome                         | Probability | Exactly | Pays   | Returns per unit | Contribution to RTP |
|---------------------------------|-------------|---------|--------|------------------|---------------------|
| Dice outrank the dealer's cards | 47.4537%    | 205/432 | 1 to 1 | 2                | 205/216             |

| Outcome           | Probability | Exactly | Pays  | Returns per unit | Contribution to RTP |
|-------------------|-------------|---------|-------|------------------|---------------------|
| Any other outcome | 52.5463%    | 227/432 | loses | 0                | 0                   |

- **Total return (RTP):** 205/216 (94.9074%), the sum of the contributions.
- **House edge:** 11/216 (5.0926%).
- **Variance** of the net result per unit staked: 46535/46656 (0.9974); **standard deviation** 0.9987.
- **Hit frequency** (the wager wins): 205/432 (47.4537%).

Reproduced by `packages/engine/src/games/mirror/mirror.test.ts` › *Mirror — the game over every draw of the dice and an infinite shoe* › adds up each bet's outcomes to its declared figures, and finds the most a round can pay.

## 4.2 Tie (side)

**Wins** when  $\text{rank}(\text{dice}) = \text{rank}(\text{cards})$ , that is the two hands hold the same two values. **Method:** exact enumeration of the production game over  $\Omega$ .

| Outcome              | Probability | Exactly | Pays    | Returns per unit | Contribution to RTP |
|----------------------|-------------|---------|---------|------------------|---------------------|
| The hands rank equal | 5.0926%     | 11/216  | 17 to 1 | 18               | 11/12               |
| Any other outcome    | 94.9074%    | 205/216 | loses   | 0                | 0                   |

- **Total return (RTP):** 11/12 (91.6667%), the sum of the contributions.
- **House edge:** 1/12 (8.3333%).
- **Variance** of the net result per unit staked: 2255/144 (15.6597); **standard deviation** 3.9572.
- **Hit frequency** (the wager wins): 11/216 (5.0926%).

Reproduced by `packages/engine/src/games/mirror/mirror.test.ts` › *Mirror — the game over every draw of the dice and an infinite shoe* › adds up each bet's outcomes to its declared figures, and finds the most a round can pay.

## 4.3 Equal Sums (side)

**Wins** when  $d_1 + d_2 = c_1 + c_2$ . **Method:** exact enumeration of the production game over  $\Omega$ .

| Outcome           | Probability | Exactly | Pays   | Returns per unit | Contribution to RTP |
|-------------------|-------------|---------|--------|------------------|---------------------|
| Same sum          | 11.2654%    | 73/648  | 7 to 1 | 8                | 73/81               |
| Any other outcome | 88.7346%    | 575/648 | loses  | 0                | 0                   |

- **Total return (RTP):** 73/81 (90.1235%), the sum of the contributions.
- **House edge:** 8/81 (9.8765%).
- **Variance** of the net result per unit staked: 41975/6561 (6.3977); **standard deviation** 2.5294.
- **Hit frequency** (the wager wins): 73/648 (11.2654%).

Reproduced by `packages/engine/src/games/mirror/mirror.test.ts` › *Mirror — the game over every draw of the dice and an infinite shoe* › adds up each bet's outcomes to its declared figures, and finds the most a round can pay.

## 4.4 Pair vs Pair (side)

**Wins** when  $d_1 = d_2$  and  $c_1 = c_2$ . **Method:** exact enumeration of the production game over  $\Omega$ .

| Outcome              | Probability | Exactly | Pays    | Returns per unit | Contribution to RTP |
|----------------------|-------------|---------|---------|------------------|---------------------|
| Both hands are pairs | 2.7778%     | 1/36    | 30 to 1 | 31               | 31/36               |
| Any other outcome    | 97.2222%    | 35/36   | loses   | 0                | 0                   |

- **Total return (RTP):** 31/36 (86.1111%), the sum of the contributions.
- **House edge:** 5/36 (13.8889%).
- **Variance** of the net result per unit staked: 33635/1296 (25.9529); **standard deviation** 5.0944.
- **Hit frequency** (the wager wins): 1/36 (2.7778%).

Reproduced by `packages/engine/src/games/mirror/mirror.test.ts` › *Mirror — the game over every draw of the dice and an infinite shoe* › adds up each bet's outcomes to its declared figures, and finds the most a round can pay.

## 4.5 Perfect Mirror (side)

**Wins** when  $d_1 = d_2 = c_1 = c_2$ . **Method:** exact enumeration of the production game over  $\Omega$ .

| Outcome                      | Probability | Exactly | Pays     | Returns per unit | Contribution to RTP |
|------------------------------|-------------|---------|----------|------------------|---------------------|
| Both hands are the same pair | 0.4630%     | 1/216   | 200 to 1 | 201              | 67/72               |
| Any other outcome            | 99.5370%    | 215/216 | loses    | 0                | 0                   |

- **Total return (RTP):** 67/72 (93.0556%), the sum of the contributions.
- **House edge:** 5/72 (6.9444%).
- **Variance** of the net result per unit staked: 965135/5184 (186.1757); **standard deviation** 13.6446.
- **Hit frequency** (the wager wins): 1/216 (0.4630%).

Reproduced by `packages/engine/src/games/mirror/mirror.test.ts` › *Mirror — the game over every draw of the dice and an infinite shoe* › adds up each bet's outcomes to its declared figures, and finds the most a round can pay.

## 4.6 Double Sixes (side, progressive)

**Wins** when  $d_1 = d_2 = c_1 = c_2 = 6$ . **Method:** exact enumeration of the production game over  $\Omega$ , with the meter at zero: the fixed pay alone.

| Outcome            | Probability | Exactly | Pays      | Returns per unit | Contribution to RTP |
|--------------------|-------------|---------|-----------|------------------|---------------------|
| Both hands are 6-6 | 0.0772%     | 1/1296  | 1000 to 1 | 1001             | 1001/1296           |

| Outcome           | Probability | Exactly   | Pays  | Returns per unit | Contribution to RTP |
|-------------------|-------------|-----------|-------|------------------|---------------------|
| Any other outcome | 99.9228%    | 1295/1296 | loses | 0                | 0                   |

- **Total return (RTP):** 1001/1296 (77.2377%), the sum of the contributions.
- **House edge:** 295/1296 (22.7623%).
- **Variance** of the net result per unit staked: 1297591295/1679616 (772.5524); **standard deviation** 27.7948.
- **Hit frequency** (the wager wins): 1/1296 (0.0772%).

With the meter at its seed, 5,000.00, a hit pays 1,200× plus the stake per unit, and one round returns 1201/1296 (92.6698%) with a standard deviation of 33.3482: the volatility the summary declares. The declared RTP, 87.2377%, counts the fixed pays and the 10% of every stake the meter pays back in the long run, the seed excluded; section five gives the meter's economics.

*Reproduced by `packages/engine/src/games/mirror/mirror.test.ts` › `Mirror` — the game over every draw of the dice and an infinite shoe › adds up each bet's outcomes to its declared figures, and finds the most a round can pay.*

## 5. Progressive analysis

| Term                                                    | Value                      |
|---------------------------------------------------------|----------------------------|
| Seed (the meter starts at it and never drops below it)  | 5,000.00                   |
| Contribution rate (of every stake, into the meter)      | 1/10 (10%)                 |
| Full-share stake (a hit takes stake ÷ it of the meter)  | 25.00                      |
| Hit chance per round                                    | 1/1296 (0.0772%)           |
| Rounds per hit, on average                              | 1296                       |
| Fixed-pay RTP                                           | 1001/1296 (77.2377%)       |
| Contribution RTP (paid back through the meter)          | 1/10 (10%)                 |
| RTP excluding the seed (declared)                       | 5653/6480 (87.2377%)       |
| RTP of one round at the seed                            | 1201/1296 (92.6698%)       |
| RTP added per cent of meter                             | 1/3240000                  |
| Break-even meter (one round returns all that is staked) | 7,375.00                   |
| Seed cost per round, at the full-share stake            | 3.86 (15.43% of the stake) |

- **The meter's rule.** Every stake on Double Sixes adds 10% of itself to the meter as the wager is accepted, kept to a millionth of a cent. A hit pays 1000 to 1 from the table's bank plus stake ÷ 25.00 of the meter, rounded down to the cent; the meter is decreased by the share paid and keeps the rest, and the house tops it back up to the seed if it fell below.
- **Return at a given meter.** A hit pays meter ÷ 25.00 per unit staked, whatever the stake, so with the meter at M cents one round returns 77.24% + M × 1/3240000: 92.67% at the seed, and all that is staked at 7,375.00, the break-even meter (8,548.91 on the 6-deck shoe).
- **The meter at a hit.** If every cycle starts at the seed, the meter at a hit averages the seed plus a cycle's contributions, seed + rate × mean stake ÷ hit chance: 5,064.80 at a mean stake of 0.50, 5,129.60 at a mean stake of 1.00, 5,648.00 at a mean stake of 5.00, 6,020.60 at a mean stake of 7.875 and 8,240.00 at a mean stake of 25.00. With stakes below the full share a hit takes only part of the meter and the rest carries over, so the meter settles higher: in the simulations of section seven, with stakes of 0.50, 1.00, 5.00 and 25.00 in turn, it averaged 7,392.57 at a hit on the infinite shoe and 7,497.68 on the 6-deck shoe.
- **The seed's cost.** At the full-share stake every hit takes the whole meter and the house re-seeds it: 3.86 a round, 15.43% of the stake. In the mixed-stake simulations the top-ups came to 1.00 and 0.97 a round, and the wager returned 99.69% and 96.57% with them.

*Reproduced by `packages/engine/src/games/mirror/mirror.test.ts` › `Mirror` — the progressive meter › prices the meter, and pays the worked example of the rules; `packages/engine/src/games/mirror/mirror.math.test.ts` › `Mirror` — Monte Carlo on an infinite shoe against the declared figures › lands every bet on its declared RTP and hit frequency; `packages/engine/src/games/mirror/six-deck.math.test.ts` › `Mirror` — the six-deck shoe in the long run › returns the exact six-deck figures, and reports the meter and the counting exposure.*

## 6. Finite-shoe effects

On the table's 6-deck shoe the second card matches the first less often than on an infinite shoe (a pair of cards is 23/143 as likely), which moves every wager. The exact figures come from running the production game over every roll and every ordered pair of cards from a full shoe (30,888 outcomes); the simulation confirms them on the real shoe, with its cut card and reshuffles.

| Wager                            | RTP, declared | RTP, six-deck shoe (exact) | RTP, six-deck shoe (observed) | Standard error |
|----------------------------------|---------------|----------------------------|-------------------------------|----------------|
| Mirror                           | 94.907%       | 95.474%                    | 95.485%                       | 0.018 pp       |
| Tie                              | 91.667%       | 91.958%                    | 91.819%                       | 0.071 pp       |
| Equal Sums                       | 90.123%       | 90.287%                    | 90.192%                       | 0.045 pp       |
| Pair vs Pair                     | 86.111%       | 83.100%                    | 82.967%                       | 0.090 pp       |
| Perfect Mirror                   | 93.056%       | 89.802%                    | 89.287%                       | 0.239 pp       |
| Double Sixes, fixed pays         | 77.238%       | 74.537%                    | 74.309%                       | 0.791 pp       |
| Double Sixes, excluding the seed | 87.238%       | 84.537%                    | 84.307%                       | 0.878 pp       |

These six-deck figures are exact for every round, not only the first after a shuffle, and a table that reshuffles before every round returns them too. *Reproduced by `packages/engine/src/games/mirror/finite-shoe.test.ts` › `Mirror` — every round of the six-deck shoe, exactly › records every bet on the six-deck shoe, and where the meter breaks even there; `packages/engine/src/games/mirror/six-deck.math.test.ts` › `Mirror` — the six-deck shoe in the long run › returns the exact six-deck figures, and reports the meter and the counting exposure.*

## 7. Simulation verification

Both runs play every wager in every round, with Double Sixes staked at 0.50, 1.00, 5.00 and 25.00 in turn and its meter live.

**On the infinite shoe,** against the declared figures:

|               |                                                                                                                                                                                                                  |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Rounds</b> | <b>124,852,059</b>                                                                                                                                                                                               |
| Generator     | xoshiro128** 1.1 (Blackman & Vigna), its state expanded from the seed by SplitMix64; a string seed is first hashed with FNV-1a (64-bit) over its UTF-8 bytes                                                     |
| Seed          | mirror/infinite-shoe                                                                                                                                                                                             |
| Cards         | an infinite shoe (every card drawn independently and uniformly from one deck of ranks 1 to 6 in 4 suits)                                                                                                         |
| Stakes        | Mirror 1.00, Tie 0.50, Equal Sums 0.50, Pair vs Pair 0.50 and Perfect Mirror 0.50, and Double Sixes as above                                                                                                     |
| Criterion     | each RTP within $\pm 0.15$ pp of the declared figure, or 3.29 of its own standard errors for the wagers too volatile for that within 200,000,000 rounds; each hit frequency within 3.29 binomial standard errors |

| Figure                           | Observed | Expected | Difference | Standard error | 95% confidence interval | Allowed         | Result |
|----------------------------------|----------|----------|------------|----------------|-------------------------|-----------------|--------|
| Mirror, RTP                      | 94.917%  | 94.907%  | +0.009 pp  | 0.009 pp       | 94.899% to 94.934%      | $\pm 0.150$ pp  | pass   |
| Tie, RTP                         | 91.642%  | 91.667%  | -0.025 pp  | 0.035 pp       | 91.573% to 91.712%      | $\pm 0.150$ pp  | pass   |
| Equal Sums, RTP                  | 90.136%  | 90.123%  | +0.012 pp  | 0.023 pp       | 90.092% to 90.180%      | $\pm 0.150$ pp  | pass   |
| Pair vs Pair, RTP                | 86.085%  | 86.111%  | -0.026 pp  | 0.046 pp       | 85.996% to 86.175%      | $\pm 0.150$ pp  | pass   |
| Perfect Mirror, RTP              | 92.917%  | 93.056%  | -0.138 pp  | 0.122 pp       | 92.678% to 93.156%      | $\pm 0.402$ pp  | pass   |
| Double Sixes, fixed pays         | 76.984%  | 77.238%  | -0.253 pp  | 0.402 pp       | 76.196% to 77.773%      | $\pm 1.323$ pp  | pass   |
| Double Sixes, excluding the seed | 86.984%  | 87.238%  | -0.253 pp  | 0.445 pp       | 86.113% to 87.856%      | $\pm 1.463$ pp  | pass   |
| Mirror, hit frequency            | 47.4584% | 47.4537% | +0.0047 pp | 0.0045 pp      | 47.4496% to 47.4671%    | $\pm 0.0147$ pp | pass   |
| Tie, hit frequency               | 5.0912%  | 5.0926%  | -0.0014 pp | 0.0020 pp      | 5.0874% to 5.0951%      | $\pm 0.0065$ pp | pass   |
| Equal Sums, hit frequency        | 11.2670% | 11.2654% | +0.0016 pp | 0.0028 pp      | 11.2614% to 11.2725%    | $\pm 0.0093$ pp | pass   |
| Pair vs Pair, hit frequency      | 2.7769%  | 2.7778%  | -0.0008 pp | 0.0015 pp      | 2.7741% to 2.7798%      | $\pm 0.0048$ pp | pass   |
| Perfect Mirror, hit frequency    | 0.4623%  | 0.4630%  | -0.0007 pp | 0.0006 pp      | 0.4611% to 0.4635%      | $\pm 0.0020$ pp | pass   |
| Double Sixes, hit frequency      | 0.0773%  | 0.0772%  | +0.0002 pp | 0.0002 pp      | 0.0768% to 0.0778%      | $\pm 0.0008$ pp | pass   |

Reproduced by `packages/engine/src/games/mirror/mirror.math.test.ts` > *Mirror — Monte Carlo on an infinite shoe against the declared figures > lands every bet on its declared RTP and hit frequency.*

On the 6-deck shoe, against the exact six-deck figures:

|               |                                                                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Rounds</b> | <b>31,213,015</b>                                                                                                                                            |
| Generator     | xoshiro128** 1.1 (Blackman & Vigna), its state expanded from the seed by SplitMix64; a string seed is first hashed with FNV-1a (64-bit) over its UTF-8 bytes |
| Seed          | mirror/six-deck-shoe                                                                                                                                         |
| Cards         | the 6-deck shoe of 144 cards, ranks 1 to 6, with its cut card after 108 cards (penetration 75%) and reshuffles, as the production game deals it              |
| Stakes        | as on the infinite shoe                                                                                                                                      |
| Criterion     | each figure within 3.29 of its own standard errors of the exact six-deck figure; the round count holds Pair vs Pair to $\pm 0.3$ pp                          |

| Figure                           | Observed | Expected | Difference | Standard error | 95% confidence interval | Allowed         | Result |
|----------------------------------|----------|----------|------------|----------------|-------------------------|-----------------|--------|
| Mirror, RTP                      | 95.485%  | 95.474%  | +0.011 pp  | 0.018 pp       | 95.450% to 95.520%      | $\pm 0.059$ pp  | pass   |
| Tie, RTP                         | 91.819%  | 91.958%  | -0.139 pp  | 0.071 pp       | 91.680% to 91.958%      | $\pm 0.233$ pp  | pass   |
| Equal Sums, RTP                  | 90.192%  | 90.287%  | -0.095 pp  | 0.045 pp       | 90.103% to 90.281%      | $\pm 0.149$ pp  | pass   |
| Pair vs Pair, RTP                | 82.967%  | 83.100%  | -0.133 pp  | 0.090 pp       | 82.792% to 83.143%      | $\pm 0.295$ pp  | pass   |
| Perfect Mirror, RTP              | 89.287%  | 89.802%  | -0.515 pp  | 0.239 pp       | 88.818% to 89.756%      | $\pm 0.787$ pp  | pass   |
| Double Sixes, fixed pays         | 74.309%  | 74.537%  | -0.228 pp  | 0.791 pp       | 72.758% to 75.859%      | $\pm 2.602$ pp  | pass   |
| Double Sixes, excluding the seed | 84.307%  | 84.537%  | -0.230 pp  | 0.878 pp       | 82.586% to 86.029%      | $\pm 2.890$ pp  | pass   |
| Mirror, hit frequency            | 47.7423% | 47.7370% | +0.0054 pp | 0.0089 pp      | 47.7248% to 47.7599%    | $\pm 0.0294$ pp | pass   |
| Tie, hit frequency               | 5.1011%  | 5.1088%  | -0.0077 pp | 0.0039 pp      | 5.0933% to 5.1088%      | $\pm 0.0130$ pp | pass   |
| Equal Sums, hit frequency        | 11.2740% | 11.2859% | -0.0119 pp | 0.0057 pp      | 11.2629% to 11.2851%    | $\pm 0.0186$ pp | pass   |
| Pair vs Pair, hit frequency      | 2.6764%  | 2.6807%  | -0.0043 pp | 0.0029 pp      | 2.6707% to 2.6820%      | $\pm 0.0095$ pp | pass   |
| Perfect Mirror, hit frequency    | 0.4442%  | 0.4468%  | -0.0026 pp | 0.0012 pp      | 0.4419% to 0.4466%      | $\pm 0.0039$ pp | pass   |
| Double Sixes, hit frequency      | 0.0740%  | 0.0745%  | -0.0004 pp | 0.0005 pp      | 0.0731% to 0.0750%      | $\pm 0.0016$ pp | pass   |

The 95% confidence interval is the observed value  $\pm 1.96$  standard errors, and at that level about one figure in twenty falls outside its interval by chance. The tests' own criterion, "Allowed" above, is wider: 3.29 standard errors, or a fixed tolerance. *Reproduced by* `packages/engine/src/games/mirror/six-deck.math.test.ts` > *Mirror — the six-deck shoe in the long run > returns the exact six-deck figures, and reports the meter and the counting exposure.*

## 8. Assumptions and limitations

**Independence.** The dice and the cards are independent: the RNG draws each die and each card separately (a live table uses separate devices), and nothing about the dice changes the shoe. Within a round the cards are dealt from one shoe, which the finite-shoe figures account for.

**Card counting.** A player who tracks the cards dealt knows the shoe's composition before every round. The 6-deck simulation computes each wager's exact expectation for that composition, round by round (for Double Sixes also with the meter as it stood):

| Wager                        | Rounds favouring a perfect counter | Counter's edge in them | Break-even bet spread |
|------------------------------|------------------------------------|------------------------|-----------------------|
| Mirror                       | 10.80%                             | 2.8%                   | 1 to 16.2             |
| Tie                          | 0.00%                              | —                      | —                     |
| Equal Sums                   | 0.00%                              | 0.7%                   | 1 to 591,198.1        |
| Pair vs Pair                 | 0.06%                              | 3.4%                   | 1 to 8,457.7          |
| Perfect Mirror               | 0.61%                              | 3.5%                   | 1 to 485.5            |
| Double Sixes                 | 11.77%                             | 22.8%                  | 1 to 10.5             |
| Double Sixes, with the meter | 41.11%                             | 26.3%                  | 1 to 1.3              |

The break-even spread is how much more a perfect counter must stake in the favourable rounds than in the others (one unit) to break even. The main wager and Double Sixes are exposed; with the meter above break-even much of the time, a counter needs almost no spread on Double Sixes. A table that reshuffles before every round, as a continuous shuffler does, removes the exposure and changes no figure of this report. *Reproduced by packages/engine/src/games/mirror/six-deck.math.test.ts > Mirror — the six-deck shoe in the long run > returns the exact six-deck figures, and reports the meter and the counting exposure.*

**Rounding.** Fixed pays are whole units per unit staked, so they are exact. A meter share is rounded down to the cent and the rest stays in the meter, to a millionth of a cent: nothing is lost, and the long-run return is unchanged.

**The meter** is modelled for one table with its own meter, as the demo runs it; a meter shared across tables changes the contributions per hit, not the rules. The seed and top-ups are the operator's cost and are excluded from the declared RTP.

**No player or dealer error** is modelled, nor any irregularity: the figures are those of the rules as the engine settles them.

**Randomness.** The analysis assumes a certified RNG: every die face and every card position equally likely and independent. The engine draws integers by rejection sampling, without modulo bias, and shuffles by Fisher–Yates in a specified order.

## 9. Appendix

**The kinds of hand**, strongest first, with their chance for the dice (and for two cards from an infinite shoe):

| Rank | Hand          | Values | Chance |
|------|---------------|--------|--------|
| 1    | PAIR 6s       | 6-6    | 1/36   |
| 2    | PAIR 5s       | 5-5    | 1/36   |
| 3    | PAIR 4s       | 4-4    | 1/36   |
| 4    | PAIR 3s       | 3-3    | 1/36   |
| 5    | PAIR 2s       | 2-2    | 1/36   |
| 6    | PAIR 1s       | 1-1    | 1/36   |
| 7    | SUM 11 HIGH 6 | 6-5    | 1/18   |
| 8    | SUM 10 HIGH 6 | 6-4    | 1/18   |
| 9    | SUM 9 HIGH 6  | 6-3    | 1/18   |
| 10   | SUM 9 HIGH 5  | 5-4    | 1/18   |
| 11   | SUM 8 HIGH 6  | 6-2    | 1/18   |
| 12   | SUM 8 HIGH 5  | 5-3    | 1/18   |
| 13   | SUM 7 HIGH 6  | 6-1    | 1/18   |
| 14   | SUM 7 HIGH 5  | 5-2    | 1/18   |
| 15   | SUM 7 HIGH 4  | 4-3    | 1/18   |
| 16   | SUM 6 HIGH 5  | 5-1    | 1/18   |
| 17   | SUM 6 HIGH 4  | 4-2    | 1/18   |
| 18   | SUM 5 HIGH 4  | 4-1    | 1/18   |
| 19   | SUM 5 HIGH 3  | 3-2    | 1/18   |
| 20   | SUM 4 HIGH 3  | 3-1    | 1/18   |
| 21   | SUM 3 HIGH 2  | 2-1    | 1/18   |

**Every dice hand against every card hand.** Rows are the dice's hand and columns the cards', both by the rank above; W marks a win for the dice (the Mirror wager wins), T a tie and a blank a loss. The cells, weighted by the two hands' chances, give the chances of section four.

| Dice \ cards | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 |
|--------------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|
| 1            | T | W | W | W | W | W | W | W | W | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  |
| 2            |   | T | W | W | W | W | W | W | W | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  |
| 3            |   |   | T | W | W | W | W | W | W | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  |
| 4            |   |   |   | T | W | W | W | W | W | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  |
| 5            |   |   |   |   | T | W | W | W | W | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  |
| 6            |   |   |   |   |   | T | W | W | W | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  |
| 7            |   |   |   |   |   |   | T | W | W | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  |
| 8            |   |   |   |   |   |   |   | T | W | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  |
| 9            |   |   |   |   |   |   |   |   | T | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  |
| 10           |   |   |   |   |   |   |   |   |   | T  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  |
| 11           |   |   |   |   |   |   |   |   |   |    | T  | W  | W  | W  | W  | W  | W  | W  | W  | W  | W  |
| 12           |   |   |   |   |   |   |   |   |   |    |    | T  | W  | W  | W  | W  | W  | W  | W  | W  | W  |
| 13           |   |   |   |   |   |   |   |   |   |    |    |    | T  | W  | W  | W  | W  | W  | W  | W  | W  |
| 14           |   |   |   |   |   |   |   |   |   |    |    |    |    | T  | W  | W  | W  | W  | W  | W  | W  |
| 15           |   |   |   |   |   |   |   |   |   |    |    |    |    |    | T  | W  | W  | W  | W  | W  | W  |
| 16           |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    | T  | W  | W  | W  | W  | W  |
| 17           |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    | T  | W  | W  | W  | W  |
| 18           |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    | T  | W  | W  | W  |
| 19           |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    | T  | W  | W  |
| 20           |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    | T  | W  |
| 21           |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    | T  |

*Reproduced by packages/engine/src/games/mirror/mirror.test.ts > Mirror — the declared table, from 36 × 24 × 24 draws > ranks the 21 kinds of hand as the rules state, and every dice hand against every card hand.*

**Records.** The tests recorded these figures ( packages/engine/src/games/mirror/results/ ):

| Record              | Test                                                                                                                                                                                                            |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| exact.json          | packages/engine/src/games/mirror/mirror.test.ts › Mirror — the game over every draw of the dice and an infinite shoe › adds up each bet's outcomes to its declared figures, and finds the most a round can pay  |
| hands.json          | packages/engine/src/games/mirror/mirror.test.ts › Mirror — the declared table, from $36 \times 24 \times 24$ draws › ranks the 21 kinds of hand as the rules state, and every dice hand against every card hand |
| infinite-shoe.json  | packages/engine/src/games/mirror/mirror.math.test.ts › Mirror — Monte Carlo on an infinite shoe against the declared figures › lands every bet on its declared RTP and hit frequency                            |
| meter.json          | packages/engine/src/games/mirror/mirror.test.ts › Mirror — the progressive meter › prices the meter, and pays the worked example of the rules                                                                   |
| six-deck-exact.json | packages/engine/src/games/mirror/finite-shoe.test.ts › Mirror — every round of the six-deck shoe, exactly › records every bet on the six-deck shoe, and where the meter breaks even there                       |
| six-deck-shoe.json  | packages/engine/src/games/mirror/six-deck.math.test.ts › Mirror — the six-deck shoe in the long run › returns the exact six-deck figures, and reports the meter and the counting exposure                       |

# Lock & Roll — Rules of Play

Lock a die, roll the other, beat the cards.

## 1. Game and document

|            |                                                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Game       | Lock & Roll, an original table game of the Roll & Deal family                                                                     |
| Version    | 0.1.0                                                                                                                             |
| Document   | Rules of Play, revision 3                                                                                                         |
| Date       | 2026-09-30                                                                                                                        |
| Author     | Jonathas Costa                                                                                                                    |
| Companion  | Math Report ( docs/math/lock-and-roll.md )                                                                                        |
| Demo table | <a href="https://jonathascosta.github.io/CasinoGames/lock-and-roll">https://jonathascosta.github.io/CasinoGames/lock-and-roll</a> |

### Revision history

| Revision | Date       | Author         | Change                                                                                                                                                                                         |
|----------|------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | 2026-09-30 | Jonathas Costa | First game sheet: rules, declared math, exact and Monte Carlo tests. The strategy computed by expected value; card counting measured.                                                          |
| 2        | 2026-09-30 | Jonathas Costa | Renamed from its working title, with English names for the game and its bets. The choices are Stand and Lock, and the re-roll costs the Lock fee.                                              |
| 3        | 2026-09-30 | Jonathas Costa | Reformatted for submission as two documents, Rules of Play and Math Report, generated from the code with every figure traced to the engine or to a test; card counting measured on every game. |

Money is written in the table's currency with two decimals, as the demo shows it (the demo plays with virtual chips).

## 2. Objective

The player rolls two dice and may, once, lock one die and re-roll the other for a fee; the dealer then deals two cards. The wager wins even money when the dice total more than the cards, and equal totals go to the house.

## 3. Equipment

- **Dice.** Two standard six-sided dice, faces 1 to 6, thrown by the player (by the round's shooter at a live table: see section nine). Their **total** runs from 2 to 12.
- **Shoe.** 6 decks, each of the A to the 6 in 4 suits (24 cards): 144 cards in all, made from standard decks with every card above the 6 removed. The ace counts 1 and every other card its number. The dealer's two cards make the dealer's total.
- **Cut card.** It goes in with 36 cards behind it, a penetration of 75%: 108 cards are dealt before it comes out. When it comes out, the round in progress is completed, and the whole shoe is shuffled before the next round.
- **Lock marker.** A marker (a padlock puck, or a marked spot on the layout) set on the die the player locks.
- **Cosmetic.** Suits never count, only values, and the order of the dealer's two cards does not matter. Which die is which matters only for the decision: the player says which one to lock.
- **Also.** The layout (section four), chips, a discard rack and, at a live table, a dice cup or shaker.

## 4. Table layout

Each player position has a single spot, for the Lock & Roll wager, with the house rules printed beside it: "Ties lose", "Lock fee: 40% of the bet", "1-1 re-rolls free". A fee area next to the spot takes the Lock fee. The felt prints the one pay:

| Spot        | Wins when                          | Pays   | A 1.00 wager wins |
|-------------|------------------------------------|--------|-------------------|
| Lock & Roll | the dice total more than the cards | 1 to 1 | 1.00              |

The demo shows this layout at <https://jonathascosta.github.io/CasinoGames/lock-and-roll>. At the top are the dealer's cards, the two totals side by side (the dice's and the cards', the higher one lit at the end) and the dice. Under the dice is the decision: Stand, and Lock with its fee once a die is locked (the dice are the lock buttons: tapping one puts a padlock on it), with a Strategy hint switch, off by default, which shows what the reference strategy would play. Below is the felt with the one wager and its printed rules, then the chips, the balance, the total wagered and the Clear, Roll and Autoplay buttons, with Rules and Paytable buttons beside the balance; an RTP monitor below the table compares the rounds played with the declared figures.

## 5. Wagers

The game has one wager and no side or progressive wager. The Lock fee is not a wager: nothing is paid on it.

### 5.1 Lock & Roll (main wager)

- **When.** Before the roll.
- **Limits.** 0.50 to 250.00.
- **Wins** when the dice's total, after the player's decision, is higher than the dealer's two cards.
- **Pays** 1 to 1.
- **Pushes** never: equal totals lose.
- **Loses** when the cards' total is equal or higher.

## 5.2 The Lock fee (not a wager)

- **What.** The price of locking one die and re-rolling the other: 40% of the wager (0.40 on a 1.00 wager), rounded up to the cent. It is a whole number of cents on any wager that is a multiple of 0.05.
- **Free on 1-1.** A roll of 1-1 re-rolls for free.
- **Never returned,** whatever the result, and nothing is paid on it: a winning wager is paid 1 to 1 on the wager alone.

## 6. Sequence of play

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1. **Wager.** The dealer invites wagers; each player places a Lock & Roll wager within the limits, and the dealer closes betting ("No more bets").
2. **Shuffle, if due.** If the cut card came out during the previous round, the dealer shuffles the whole shoe, has it cut and places the cut card with 36 cards behind it before the round begins.
3. **Roll.** The player rolls both dice. The dealer announces the roll and its total.
4. **Decision, once.** Before any card is dealt, the player chooses:
  - **Stand:** keep the roll; or
  - **Lock:** name one die to keep (the dealer sets the lock marker on it), place the Lock fee, 40% of the wager, in the fee area (nothing on 1-1), and re-roll the other die once. The re-rolled die is final.

There is one decision per round. A player who has not decided when the dealer calls time stands: no fee is taken without the player's choice.

5. **Collect the fee.** The dealer takes each Lock fee as the lock is made, before the re-roll.
6. **Re-roll.** The locking player re-rolls the unlocked die; the dealer announces the new total.
7. **Deal.** The dealer deals two cards from the shoe, face up, and announces their total.
8. **Compare.** The dealer compares the dice's total with the cards' total (section seven).
9. **Settle.** The dealer settles the wager (section eight), clears the cards and returns the dice.

## 7. Comparing the totals

---

There are no hands to rank: the dice's total, after the decision, is compared with the dealer's two cards.

- The dice's total is the sum of the two dice as they stand after the decision: the roll if the player stood, the locked die and the re-rolled die otherwise.
- The dealer's total is the sum of the two cards, the ace counting 1.
- A higher dice total wins; an equal or lower one loses. There is no push.

Worked examples:

- Wager 1.00, roll 6-1. The player locks the 6 and re-rolls the 1 for 0.40; it lands on 4, for 10; the dealer deals a 5 and a 3, for 8: the dice are higher and the wager wins 1.00. The round nets +0.60 for the player, the fee included.
- Wager 1.00, roll 1-1. The player locks the 1 and re-rolls the 1, free on 1-1; it lands on 5, for 6; the dealer deals a 3 and a 3, for 6: a tie, which the house wins. The round nets -1.00 for the player.
- Wager 1.00, roll 6-2. The player stands on 8; the dealer deals a 6 and a 5, for 11: the cards are higher and the wager loses. The round nets -1.00 for the player.

## 8. Settlement procedure

---

- **The Lock fee** is collected when the player locks, before the re-roll, and is never returned: it is already the house's when the cards are dealt.
- **The wager** is settled once the cards are dealt: a losing wager is collected; a winning one is paid 1 to 1 and keeps its stake. Nothing is paid on the fee.
- With several players, the dealer collects every losing wager first, then pays the winners.
- There is no progressive meter.

## 9. Physical and live-dealer adaptation

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- **Dice.** The player throws from a dice cup or by hand, across the layout. At a table with several players, one roll (from a shooter in turn, or an automatic shaker) serves every seat; each seat then decides on its own. The re-roll becomes one more die, rolled once for every seat that locked: each of those seats keeps the die it locked and adds the new die.
- **Timing.** The dealer calls for decisions after the roll and closes them before the re-roll; seats that have not decided stand. The cards are dealt only after the re-roll.
- **Calls.** The dealer announces the roll, the decisions and fees taken, the re-rolled die, the cards' total and the result.
- **Shoe.** Prepare 6 decks of A to 6 (144 cards), shuffle, offer the cut and place the cut card with 36 cards behind it. With two cards a round, a shoe deals 54 rounds.
- **Card counting.** The cards are dealt face up, and the wager can be counted at this penetration: the Math Report measures by how much. A live table should use a continuous shuffling machine, or shuffle much earlier.
- **The RNG version** draws the dice, the re-roll and the cards from a certified random number generator, deals from a virtual shoe with the same cut card (or reshuffles it every round) and settles at once. The player taps a die to lock it; if the table closes before the player decides, the round finishes as a Stand. The optional Strategy hint and autoplay follow the reference strategy. It has no physical irregularities. Every rule, limit and pay is the same.

## 10. Irregularities

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- **Die off the table.** A die that leaves the layout (off the table, into the chip rail or the dealer's bank) is no roll: the dealer calls it, checks the dice and the shooter rolls again (on the re-roll, only the re-rolled die: the locked die stays). Wagers stay as they are.
- **Cocked die.** A die that comes to rest tilted, against a chip, the rail, a card or the other die, so that no face is plainly uppermost, is no roll: the dice are rolled again (on the re-roll, only the re-rolled die: the locked die stays). A die lying flat on a chip is read as it lies.
- **Dice thrown too early.** Dice thrown before the dealer closes betting are no roll.
- **Misread result.** The dice and cards as they lie decide the round: a misread call is corrected before the round is settled, and a settlement error is corrected before the next round begins.
- **Exposed card.** A card exposed before the decisions are made would inform them: it is burned to the discards, the decisions go ahead, and the round is dealt from the next cards. A card exposed after the decisions is burned too.
- **Mis-deal.** Only the first two cards dealt after the re-roll count; a card dealt before the decisions or the re-roll, or any further card, is burned.
- **Wrong die re-rolled.** If the locked die is thrown instead of the other, both dice are set back as they were before the decision and the right die is re-rolled; the fee stands.
- **Cards short of the cut card.** It cannot happen as the shoe is prepared: a round uses at most two cards, and 36 cards sit behind the cut card. If the shoe were nonetheless to run out during a round (a cut card placed too deep, cards missing), the dealer shuffles the discards of earlier rounds, never the cards of the round in progress, and completes the round from them; the whole shoe is then shuffled before the next round. This is what the RNG version does.

## To confirm

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These rulings are standard casino practice, not rules the game's code decides. Confirm each one, or replace it with the house's own, before a live table opens.

- Settlement order: losing wagers collected first, then the winners paid, main wager first.
- Dice off the table and cocked dice are no roll, rather than read where they lie.
- At a table with several players, one shooter per round, in turn (or an automatic shaker), whose roll settles every seat.
- No card is burned after the shuffle, as in the RNG version; a burn card would not change the rules.
- Undecided seats stand when the dealer calls time, and pay no fee.
- At a multi-seat table, one extra die, rolled once, serves every seat that locked.
- A card exposed before the decisions is burned and the decisions go ahead.
- A die thrown by mistake is set back and the right die re-rolled, the fee standing.
- Counting: a continuous shuffler, or a much earlier shuffle, at a live table.

# Lock & Roll — Math Report

Lock a die, roll the other, beat the cards.

## 1. Cover

|           |                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|
| Game      | Lock & Roll, an original table game of the Roll & Deal family                                                                    |
| Version   | 0.1.0                                                                                                                            |
| Document  | Math Report, revision 3                                                                                                          |
| Date      | 2026-09-30                                                                                                                       |
| Author    | Jonathas Costa                                                                                                                   |
| Code      | commit 47bab22a941c983cd3e18fd6989ebb7a1e6479b5 (2026-09-30): the last change to the engine, its tests or their recorded results |
| Figures   | docs/results.json, written with this report by tools/generate-docs.ts                                                            |
| Companion | Rules of Play ( docs/rules/lock-and-roll.md )                                                                                    |

### Revision history

| Revision | Date       | Author         | Change                                                                                                                                                                                         |
|----------|------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | 2026-09-30 | Jonathas Costa | First game sheet: rules, declared math, exact and Monte Carlo tests. The strategy computed by expected value; card counting measured.                                                          |
| 2        | 2026-09-30 | Jonathas Costa | Renamed from its working title, with English names for the game and its bets. The choices are Stand and Lock, and the re-roll costs the Lock fee.                                              |
| 3        | 2026-09-30 | Jonathas Costa | Reformatted for submission as two documents, Rules of Play and Math Report, generated from the code with every figure traced to the engine or to a test; card counting measured on every game. |

**How to read this report.** Every figure is read from `docs/results.json`, which the generator builds from the game's `mathSummary()` (the declared figures) and from the figures its tests record ( `packages/engine/src/games/lock-and-roll/results/` ); the generator fails if a document holds a number that is not in it. Each table names the test that reproduces it, as file > suite > test: `pnpm test` runs the exact tests and `pnpm test:math` the simulations, and each fails if it no longer reproduces its recorded figures. Returns are per unit staked, the stake included; exact figures are fractions, rounded percentages beside them.

## 2. Summary

| Wager              | RTP      | House edge | Hit frequency | SD per unit | Max payout | Max exposure at max bet |
|--------------------|----------|------------|---------------|-------------|------------|-------------------------|
| Lock & Roll (main) | 95.9311% | 4.0689%    | 56.8544%      | 1.0341      | 1×         | 250.00 (on 250.00)      |

- **RTP** and **house edge** are per unit staked, pushes included. **Hit frequency** is the chance that the wager wins in a round. **SD per unit** is the standard deviation of the net result per unit staked.
- **Max payout** is the largest net win per unit staked, the stake being returned on top. **Max exposure at max bet** is the most a single wager at its maximum can win in a round, found by running the real game over every outcome at the table maximums; all the wagers of one position together can win at most 250.00 in a round.
- The figures assume the **optimal strategy** of section four, and count the Lock fee against the return: a fee is never returned and is not a stake, so  $RTP = (\text{payouts} - \text{fees}) \div \text{stakes}$ . Measured instead against everything the player pays, the wager and the fees, the loss is the element of risk, 791/22896 (3.45%).
- The most a round can cost the player is 350.00 at the maximum wager: the wager and the Lock fee.

Reproduced by `packages/engine/src/games/lock-and-roll/lock-and-roll.test.ts` > *Lock & Roll — the declared figures, exactly, on the game > breaks the return down by decision and result, and prices the strategies a player might follow.* The declared figures are `mathSummary()` (bets.ts); the test holds them equal to the enumeration.

## 3. Game model

- **Sample space.** The dice ( $d_1$ ,  $d_2$ ) are uniform on 1 to 6: 36 rolls. The player then stands, or locks one die and re-rolls the other once: a face  $r$  uniform on 1 to 6 (6 outcomes). The dealer's two cards ( $c_1$ ,  $c_2$ ) follow. The wager wins when the dice's final total exceeds  $c_1 + c_2$ . The exact test enumerates the production game with the reference strategy over every roll, re-roll and ordered pair of cards from one deck's 24 cards: 69,696 outcomes.
- **Card supply.** The declared figures take the two cards independent and uniform over the values: an infinite shoe, as an RNG game that draws each card independently deals them. On the table's 6-deck shoe the two cards of any round are a uniform draw of two cards from the full shoe (two cards a round and 54 rounds a shoe, whatever the dice and the decisions); section six gives those figures exactly.
- **Decision model.** After the roll the player knows the dice and chooses Stand, or Lock one die for the fee of 2/5 (40%) of the wager (free on 1-1). Each choice is valued by its expected net result per unit: standing on a total  $s$  is worth  $P(c_1 + c_2 < s) - P(c_1 + c_2 \geq s)$ , and locking a die of value  $k$  is worth the average over the re-rolled face of standing on  $k + r$ , less the fee. The **optimal strategy** takes the choice of highest value on each roll, standing when no lock is strictly better. The declared figures assume it; section four prices other strategies.

## 4. Per-wager analysis

### 4.1 Lock & Roll (main)

**Wins** when the final dice total exceeds  $c_1 + c_2$ ; a tie loses. **Method:** exact enumeration of the production game with the reference strategy over  $\Omega$ ; the strategy itself is derived by a second test from scratch, by expected value.

The return by what the player did and how the wager ended (the fee counted in the net):

| Decision | Result | Probability | Exactly  | Net per unit | Contribution to the net |
|----------|--------|-------------|----------|--------------|-------------------------|
| Stand    | wins   | 34.1821%    | 443/1296 | +1.00        | 443/1296                |

| Decision           | Result | Probability | Exactly  | Net per unit | Contribution to the net |
|--------------------|--------|-------------|----------|--------------|-------------------------|
| Stand              | loses  | 18.5957%    | 241/1296 | −1.00        | −241/1296               |
| Lock (fee paid)    | wins   | 22.2222%    | 2/9      | +0.60        | 2/15                    |
| Lock (fee paid)    | loses  | 22.2222%    | 2/9      | −1.40        | −14/45                  |
| Lock (free on 1-1) | wins   | 0.4501%     | 35/7776  | +1.00        | 35/7776                 |
| Lock (free on 1-1) | loses  | 2.3277%     | 181/7776 | −1.00        | −181/7776               |

- **Total return (RTP):** 18649/19440 (95.9311%), the sum of the contributions.
- **House edge:** 791/19440 (4.0689%).
- **Variance** of the net result per unit staked: 16166471/15116544 (1.0695); **standard deviation** 1.0341.
- **Hit frequency** (the wager wins): 4421/7776 (56.8544%).
- **Lock:** in 17/36 (47.22%) of rounds; the fee is paid in 4/9 (44.44%) of rounds, 8/45 (17.78%) of the wager on average.

Reproduced by `packages/engine/src/games/lock-and-roll/lock-and-roll.test.ts` › *Lock & Roll — the declared figures, exactly, on the game* › breaks the return down by decision and result, and prices the strategies a player might follow.

**The decision table.** Each value is the expected net result per unit of the wager, the fee included; the dice are listed higher first, and on a double both locks are worth the same.

| Roll | Chance | Stand   | Lock the higher die | Lock the lower die | Best play              | Margin  |
|------|--------|---------|---------------------|--------------------|------------------------|---------|
| 1-1  | 1/36   | −1.0000 | −0.6759             | −0.6759            | Lock either die (free) | +0.3241 |
| 2-1  | 1/18   | −0.9444 | −0.8815             | −1.0759            | Lock the higher die    | +0.0630 |
| 3-1  | 1/18   | −0.8333 | −0.6500             | −1.0759            | Lock the higher die    | +0.1833 |
| 4-1  | 1/18   | −0.6667 | −0.4000             | −1.0759            | Lock the higher die    | +0.2667 |
| 5-1  | 1/18   | −0.4444 | −0.1500             | −1.0759            | Lock the higher die    | +0.2944 |
| 6-1  | 1/18   | −0.1667 | +0.0815             | −1.0759            | Lock the higher die    | +0.2481 |
| 2-2  | 1/36   | −0.8333 | −0.8815             | −0.8815            | Stand                  | +0.0481 |
| 3-2  | 1/18   | −0.6667 | −0.6500             | −0.8815            | Lock the higher die    | +0.0167 |
| 4-2  | 1/18   | −0.4444 | −0.4000             | −0.8815            | Lock the higher die    | +0.0444 |
| 5-2  | 1/18   | −0.1667 | −0.1500             | −0.8815            | Lock the higher die    | +0.0167 |
| 6-2  | 1/18   | +0.1667 | +0.0815             | −0.8815            | Stand                  | +0.0852 |
| 3-3  | 1/36   | −0.4444 | −0.6500             | −0.6500            | Stand                  | +0.2056 |
| 4-3  | 1/18   | −0.1667 | −0.4000             | −0.6500            | Stand                  | +0.2333 |
| 5-3  | 1/18   | +0.1667 | −0.1500             | −0.6500            | Stand                  | +0.3167 |
| 6-3  | 1/18   | +0.4444 | +0.0815             | −0.6500            | Stand                  | +0.3630 |
| 4-4  | 1/36   | +0.1667 | −0.4000             | −0.4000            | Stand                  | +0.5667 |
| 5-4  | 1/18   | +0.4444 | −0.1500             | −0.4000            | Stand                  | +0.5944 |
| 6-4  | 1/18   | +0.6667 | +0.0815             | −0.4000            | Stand                  | +0.5852 |
| 5-5  | 1/36   | +0.6667 | −0.1500             | −0.1500            | Stand                  | +0.8167 |
| 6-5  | 1/18   | +0.8333 | +0.0815             | −0.1500            | Stand                  | +0.7519 |
| 6-6  | 1/36   | +0.9444 | +0.0815             | +0.0815            | Stand                  | +0.8630 |

**The optimal strategy.** Lock the higher die and re-roll the lower on 1-1, 2-1, 3-1, 4-1, 5-1, 6-1, 3-2, 4-2 and 5-2; stand on the other rolls. Locking the lower die is never best: the higher the die kept, the better the re-roll. The closest calls are decided by 1/60 of the wager. The strategy locks in 17/36 (47.22%) of rounds.

**With and without the free 1-1.** The free re-roll is worth 35/3888 (0.90%) of the wager to the player. Without it, 1-1 stands (a sure loss beats paying the fee for the re-roll), the strategy locks in 4/9 (44.44%) of rounds, and:

| Rules                | RTP                    | House edge          | Hit frequency | SD per unit |
|----------------------|------------------------|---------------------|---------------|-------------|
| As played, 1-1 free  | 18649/19440 (95.9311%) | 791/19440 (4.0689%) | 56.854%       | 1.0341      |
| Without the free 1-1 | 3079/3240 (95.0309%)   | 161/3240 (4.9691%)  | 56.404%       | 1.0338      |

**Other strategies.** The declared edge assumes optimal play; a player who plays otherwise gives the house more:

| Strategy                                                   | RTP                  | House edge          | Hit frequency | Average fee |
|------------------------------------------------------------|----------------------|---------------------|---------------|-------------|
| Optimal (the reference strategy, declared)                 | 18649/19440 (95.93%) | 791/19440 (4.07%)   | 56.85%        | 17.78%      |
| Never lock                                                 | 575/648 (88.73%)     | 73/648 (11.27%)     | 44.37%        | 0.00%       |
| Always lock the lower die (re-roll the higher every round) | 1049/3888 (26.98%)   | 2839/3888 (73.02%)  | 32.93%        | 38.89%      |
| Always lock the higher die (re-roll the lower every round) | 2827/3888 (72.71%)   | 1061/3888 (27.29%)  | 55.80%        | 38.89%      |
| Lock when the strategy does, but the lower die             | 13439/19440 (69.13%) | 6001/19440 (30.87%) | 43.45%        | 17.78%      |

Reproduced by `packages/engine/src/games/lock-and-roll/strategy.test.ts` › *Lock & Roll — the best choice on each of the 21 rolls, by expected value* › values standing and locking either die on each of the 21 rolls, as `strategy.ts` does; `packages/engine/src/games/lock-and-roll/lock-and-roll.test.ts` › *Lock & Roll — the declared figures, exactly, on the game* › breaks the return down by decision and result, and prices the strategies a player might follow.

## 5. Progressive analysis

Not applicable: Lock & Roll has no progressive wager.

## 6. Finite-shoe effects

On the table's 6-deck shoe the second card matches the first less often, which moves the dealer's totals toward the middle and helps the dice a little. The exact figures come from running the production game with the reference strategy over every roll, re-roll and ordered pair of cards from a full shoe (103,818 outcomes); the strategy is the same on both shoes, and the simulation confirms the figure on the real shoe, with its

cut card and reshuffles.

| Figure          | Declared (infinite shoe) | Six-deck shoe, exact    | Six-deck shoe, observed |
|-----------------|--------------------------|-------------------------|-------------------------|
| RTP             | 18649/19440 (95.931%)    | 148219/154440 (95.972%) | 95.861% ± 0.046 pp      |
| House edge      | 791/19440 (4.069%)       | 6221/154440 (4.028%)    |                         |
| Hit frequency   | 4421/7776 (56.854%)      | 35135/61776 (56.875%)   | 56.823%                 |
| Element of risk | 791/22896 (3.455%)       | 6221/181896 (3.420%)    |                         |

The six-deck shoe returns +0.041 pp more than declared. *Reproduced by* `packages/engine/src/games/lock-and-roll/finite-shoe.test.ts` › *Lock & Roll — every round of the six-deck shoe, exactly* › *records the six-deck figures of the reference strategy*; `packages/engine/src/games/lock-and-roll/six-deck.math.test.ts` › *Lock & Roll — the six-deck shoe in the long run* › *returns the exact six-deck figures, and reports the counting exposure*.

## 7. Simulation verification

Both runs play the production game with the reference strategy deciding every round, the same bot the demo's autoplay uses.

**On the infinite shoe**, against the declared figures:

| Rounds    | 5,144,842                                                                                                                                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Generator | xoshiro128** 1.1 (Blackman & Vigna), its state expanded from the seed by SplitMix64; a string seed is first hashed with FNV-1a (64-bit) over its UTF-8 bytes |
| Seed      | lock-and-roll/infinite-shoe                                                                                                                                  |
| Cards     | an infinite shoe (every card drawn independently and uniformly from one deck of ranks 1 to 6 in 4 suits)                                                     |
| Stakes    | 1.00 a round                                                                                                                                                 |
| Criterion | the RTP within ±0.15 pp of the declared figure (3.29 standard errors at this round count), and each frequency within 3.29 binomial standard errors           |

| Figure            | Observed | Expected | Difference | Standard error | 95% confidence interval | Allowed   | Result |
|-------------------|----------|----------|------------|----------------|-------------------------|-----------|--------|
| RTP               | 95.874%  | 95.931%  | −0.057 pp  | 0.046 pp       | 95.785% to 95.963%      | ±0.150 pp | pass   |
| Win frequency     | 56.835%  | 56.854%  | −0.019 pp  | 0.022 pp       | 56.792% to 56.878%      | ±0.072 pp | pass   |
| Lock (re-rolls)   | 47.270%  | 47.222%  | +0.048 pp  | 0.022 pp       | 47.227% to 47.314%      | ±0.072 pp | pass   |
| Free 1-1 re-rolls | 2.779%   | 2.778%   | +0.002 pp  | 0.007 pp       | 2.765% to 2.794%        | ±0.024 pp | pass   |
| Fee paid          | 44.491%  | 44.444%  | +0.047 pp  | 0.022 pp       | 44.448% to 44.534%      | ±0.072 pp | pass   |

*Reproduced by* `packages/engine/src/games/lock-and-roll/lock-and-roll.math.test.ts` › *Lock & Roll — Monte Carlo on an infinite shoe against the declared figures* › *lands on the declared RTP, win frequency, re-roll and fee frequencies*.

**On the 6-deck shoe**, against the exact six-deck figures:

| Rounds    | 5,144,842                                                                                                                                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Generator | xoshiro128** 1.1 (Blackman & Vigna), its state expanded from the seed by SplitMix64; a string seed is first hashed with FNV-1a (64-bit) over its UTF-8 bytes |
| Seed      | lock-and-roll/six-deck-shoe                                                                                                                                  |
| Cards     | the 6-deck shoe of 144 cards, ranks 1 to 6, with its cut card after 108 cards (penetration 75%) and reshuffles, as the production game deals it              |
| Stakes    | 1.00 a round; 95,275 shuffles                                                                                                                                |
| Criterion | the RTP within ±0.15 pp and 3.29 standard errors of the exact six-deck figure, the win frequency within 3.29 binomial standard errors                        |

| Figure        | Observed | Expected | Difference | Standard error | 95% confidence interval | Allowed   | Result |
|---------------|----------|----------|------------|----------------|-------------------------|-----------|--------|
| RTP           | 95.861%  | 95.972%  | −0.111 pp  | 0.046 pp       | 95.771% to 95.950%      | ±0.150 pp | pass   |
| Win frequency | 56.823%  | 56.875%  | −0.051 pp  | 0.022 pp       | 56.781% to 56.866%      | ±0.072 pp | pass   |

The 95% confidence interval is the observed value ± 1.96 standard errors, and at that level about one figure in twenty falls outside its interval by chance. The tests' own criterion, "Allowed" above, is wider: 3.29 standard errors, or a fixed tolerance. The six-deck RTP lies 2.4 standard errors from its exact figure: outside the 95% interval, inside the test's 3.29. *Reproduced by* `packages/engine/src/games/lock-and-roll/six-deck.math.test.ts` › *Lock & Roll — the six-deck shoe in the long run* › *returns the exact six-deck figures, and reports the counting exposure*.

## 8. Assumptions and limitations

**Independence.** The dice and the cards are independent: the RNG draws each die and each card separately (a live table uses separate devices), and nothing about the dice changes the shoe. Within a round the cards are dealt from one shoe, which the finite-shoe figures account for.

**Optimal play.** The declared figures assume the optimal strategy; section four gives the return of other strategies. A player who never locks gives the house 73/648 (11.27%) instead of 791/19440 (4.07%).

**Card counting.** The cards are dealt face up, so a player who tracks them knows the shoe's composition before every round. The 6-deck simulation computes the exact expectation for that composition, round by round, playing the reference strategy and deciding with the count:

| Wager                   | Rounds favouring a perfect counter | Counter's edge in them | Break-even bet spread |
|-------------------------|------------------------------------|------------------------|-----------------------|
| The reference strategy  | 21.07%                             | 4.6%                   | 1 to 5.1              |
| Deciding with the count | 21.09%                             | 4.7%                   | 1 to 5.0              |

The break-even spread is how much more a perfect counter must stake in the favourable rounds than in the others (one unit) to break even. The wager is exposed at the table's penetration; adjusting the decisions to the count adds almost nothing. A table that reshuffles before every round, as a continuous shuffler does, removes the exposure; its figures are the six-deck figures of section six. *Reproduced by* `packages/engine/src/games/lock-and-roll/six-deck.math.test.ts` › *Lock & Roll — the six-deck shoe in the long run* › *returns the exact six-deck figures, and reports the counting exposure*.

**Rounding.** The wager pays even money, exactly. The Lock fee is rounded up to the cent, which only matters on wagers that are not a multiple of 0.05; the figures assume the exact fee.

**Timing.** A player who does not decide stands (the RTP of standing on every roll is in section four's table of strategies); no fee is taken without a decision.

**Randomness.** The analysis assumes a certified RNG: every die face and every card position equally likely and independent. The engine draws integers by rejection sampling, without modulo bias, and shuffles by Fisher-Yates in a specified order.

## 9. Appendix

The full enumeration of 69,696 outcomes does not fit a page; the tests hold it. The decision table of section four, exactly:

| Roll | Stand  | Lock the higher die | Lock the lower die | Margin  |
|------|--------|---------------------|--------------------|---------|
| 1-1  | -1     | -73/108             | -73/108            | 35/108  |
| 2-1  | -17/18 | -119/135            | -581/540           | 17/270  |
| 3-1  | -5/6   | -13/20              | -581/540           | 11/60   |
| 4-1  | -2/3   | -2/5                | -581/540           | 4/15    |
| 5-1  | -4/9   | -3/20               | -581/540           | 53/180  |
| 6-1  | -1/6   | 11/135              | -581/540           | 67/270  |
| 2-2  | -5/6   | -119/135            | -119/135           | 13/270  |
| 3-2  | -2/3   | -13/20              | -119/135           | 1/60    |
| 4-2  | -4/9   | -2/5                | -119/135           | 2/45    |
| 5-2  | -1/6   | -3/20               | -119/135           | 1/60    |
| 6-2  | 1/6    | 11/135              | -119/135           | 23/270  |
| 3-3  | -4/9   | -13/20              | -13/20             | 37/180  |
| 4-3  | -1/6   | -2/5                | -13/20             | 7/30    |
| 5-3  | 1/6    | -3/20               | -13/20             | 19/60   |
| 6-3  | 4/9    | 11/135              | -13/20             | 49/135  |
| 4-4  | 1/6    | -2/5                | -2/5               | 17/30   |
| 5-4  | 4/9    | -3/20               | -2/5               | 107/180 |
| 6-4  | 2/3    | 11/135              | -2/5               | 79/135  |
| 5-5  | 2/3    | -3/20               | -3/20              | 49/60   |
| 6-5  | 5/6    | 11/135              | -3/20              | 203/270 |
| 6-6  | 17/18  | 11/135              | 11/135             | 233/270 |

Reproduced by `packages/engine/src/games/lock-and-roll/strategy.test.ts` › *Lock & Roll — the best choice on each of the 21 rolls, by expected value › values standing and locking either die on each of the 21 rolls, as strategy.ts does.*

**Records.** The tests recorded these figures ( `packages/engine/src/games/lock-and-roll/results/` ):

| Record                           | Test                                                                                                                                                                                                                                   |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>exact.json</code>          | <code>packages/engine/src/games/lock-and-roll/lock-and-roll.test.ts</code> › Lock & Roll — the declared figures, exactly, on the game › breaks the return down by decision and result, and prices the strategies a player might follow |
| <code>infinite-shoe.json</code>  | <code>packages/engine/src/games/lock-and-roll/lock-and-roll.math.test.ts</code> › Lock & Roll — Monte Carlo on an infinite shoe against the declared figures › lands on the declared RTP, win frequency, re-roll and fee frequencies   |
| <code>six-deck-exact.json</code> | <code>packages/engine/src/games/lock-and-roll/finite-shoe.test.ts</code> › Lock & Roll — every round of the six-deck shoe, exactly › records the six-deck figures of the reference strategy                                            |
| <code>six-deck-shoe.json</code>  | <code>packages/engine/src/games/lock-and-roll/six-deck.math.test.ts</code> › Lock & Roll — the six-deck shoe in the long run › returns the exact six-deck figures, and reports the counting exposure                                   |
| <code>strategy.json</code>       | <code>packages/engine/src/games/lock-and-roll/strategy.test.ts</code> › Lock & Roll — the best choice on each of the 21 rolls, by expected value › values standing and locking either die on each of the 21 rolls, as strategy.ts does |